EXACTLY THREE THETA MATES FOR A FAMILY OF RANK-FOUR LATTICES

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Abstract. A tensor construction over the Eisenstein integers attaches to each positive definite integral binary form $M = (a, b, c)$ three even lattices L_0, L_1, L_2 of rank four with equal theta series; for $0 < 2b < a < c$ they are pairwise non-isometric. We prove that for these parameters the theta class of L_0 contains no further isometry class: every positive definite lattice of rank four with the theta series of L_0 is isometric to one of L_0, L_1, L_2. The proof treats $D = ac - b^2 \not\equiv 2 \pmod 3$ by congruence reductions, ambient rigidity, descents and ascents, and $D \equiv 2 \pmod 3$ by a decomposition of the normalized parameter region into chambers, each reduced to a finite symbolic branch universe whose branches are settled by exact certificates and exact isometry tests. Several steps are computer-assisted; each computational dependency is pinned to its data and has a declared verification route. We prove no upper bound for rank-four theta fibres in general.

1. Introduction

Two lattices with the same theta series need not be isometric. In rank three a positive definite lattice is determined by its theta series [5]; in rank four Conway and Sloane gave a four-parameter family of theta-equal pairs [2], proved non-isometric by Cerviño and Hein [1]. Theta-equal, non-isometric lattices of rank four give isospectral, non-isometric flat tori, through their dual lattices. The maximal number of pairwise non-isometric, mutually isospectral flat tori of dimension four is denoted $\flat_4$.

A tensor construction over the Eisenstein integers gives, for each positive definite integral binary form $M = (a, b, c)$, three even lattices L_0, L_1, L_2 of rank four with equal theta series. For generic M $(0 < 2b < a < c)$ they are pairwise non-isometric (Section 2). In particular $\flat_4 \geq 3$, contradicting the conjecture as stated in arXiv:2412.16709v1 [3, Outlook]. We prove no upper bound for $\flat_4$.

This paper answers the natural next question for this family: does the theta class of L_0 contain any further isometry classes?

Theorem 1.1 (Main theorem)**.** *Let a, b, c be integers with $0 < 2b < a < c$. Every positive definite lattice of rank four with the theta series of $L_0(a, b, c)$ is isometric to exactly one of L_0, L_1, L_2.*

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Scope.

- The theorem has no bound on a, b, c or $D = ac - b^2$, and no residue or coprimality condition.
- It concerns this generic constructed family only. It does not bound the size of rank-four theta fibres in general, and it does not treat the non-generic parameters $b = 0$, $2b = a$ or $a = c$.

Structure of the proof.

- Section 3: primitive seeds with $D \not\equiv 2 \pmod 3$, by congruence reduction, ambient rigidity, descents and ascents.
- Sections 4–9: primitive seeds with $D \equiv 2 \pmod 3$, split into chambers in the normalized coordinates $(\beta, \gamma) = (b/a, c/a)$, each with its walls assigned; finitely many seeds with $a \leq 10$ are closed by the finite register (Remark 4.2).
- Section 10: common scalings, and the assembly of the main theorem.

Computer assistance. Several steps are computer-assisted:

- the exact finite reduction in the proof of ambient rigidity (Theorem 3.1);
- the within-seed classification certificate used for $D \equiv 1 \pmod 3$ (Theorem 3.2);
- the residue certificate for $D \equiv 0, 3 \pmod 9$;
- the rigidity statement (R_Z) for $D \equiv 6 \pmod 9$;
- the chamber exclusions for $D \equiv 2 \pmod 3$;
- the finite register of small seeds for $D \equiv 2 \pmod 3$ below the tail (Remark 4.2);
- the streamer recoveries, coset bands and certificate ladder of the tail theorem (Theorem 4.3).

Each chamber exclusion reduces to a finite symbolic branch universe. Each branch is excluded by exact certificates, identified with an explicit integral basis of a known lattice, or settled by exact theta differences and exact isometry tests. Agreement of finitely many theta coefficients is never used as a positive identification.

Section 11 states exactly:

- what has been re-executed independently;
- what has been audited in full, and what has been sampled;
- how to reproduce each step from the accompanying data.

Appendix A gives every step a stable identifier, its status and its verification route; the machine-readable supplement (`SUPPLEMENT_MANIFEST.json` and the dependency file it names) pins its data, hashes and jobs.

Authorship and contributions. This work is released by bigjokker, under the license of the accompanying repository. It was carried out with AI assistance. Claude sessions (Claude Opus 5.5, with earlier contributions from Claude Fable 5.1 sessions) designed and ran the proofs and computations, including the chamber searches and the C3 graph-one theorem; Grok

and Grok Bot supplied independent implementations and literature checks; Codex sessions supplied package audits, the solver-free certificate checker, proof notes and the final reviews. These are AI-assisted reviews with independent computational replication, not peer review. No publication, priority or novelty claim is made. PARI/GP [4] was used for exact theta series and isometry tests.

2. The construction and notation

Lattices and theta series. A lattice is a free $\mathbb{Z}$-module of finite rank with a positive definite symmetric bilinear form B. Gram matrices are matrices of B, and $Q(x) = B(x,x)/2$. A lattice is *even* if Q is $\mathbb{Z}$-valued; all lattices here are even. The theta series is

$$\theta_L = \sum_{x\in L} q^{Q(x)} = \sum_{n\geq 0} r_L(n)\, q^n, \qquad r_L(n) = \#\{x \in L : Q(x) = n\}.$$

A *theta mate* of L is a lattice K with $\theta_K = \theta_L$. Isometry means integral isometry.

The seed. Let $E = \mathbb{Z}[\omega]$ with $\omega^2 = \omega - 1$. Write N and Tr for the norm and trace. Let $M = \left(\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right)$ be a positive definite integral binary form, and $D = ac - b^2$. On E^2 put

$$Q(z_1, z_2) = a\, N(z_1) + c\, N(z_2) + b\, \mathrm{Tr}(\bar{z}_1 z_2).$$

In the basis $(1,0), (\omega,0), (0,1), (0,\omega)$ its Gram matrix is $M \otimes A_2$, of determinant $9D^2$. We call M, or the triple (a,b,c), a *seed*. It is *generic* if $0 < 2b < a < c$, and *primitive* if $\gcd(a,b,c) = 1$.

The three lattices. Put

$$\Phi_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \Phi_1 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad \Phi_2 = \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix},$$

acting on the coordinate column of z_1 in the basis $(1, \omega)$. Define

$$L_i = L_i(M) = \{(z_1, z_2) \in E^2 :\ z_2 \equiv \Phi_i z_1 \pmod 3\}, \qquad i = 0, 1, 2.$$

Each L_i has index 9 in E^2 and determinant $(27D)^2$. Its residue mod 3 is the *graph plane* $P_i = \{(\bar{z}, \Phi_i \bar{z})\} \subset \mathbb{F}_3^4$, and we refer to i as the *graph*. Writing $z_1 = x + y\omega$ and $z_2 = u + w\omega$, the congruence for L_1 reads $u \equiv x + y$, $w \equiv x \pmod 3$. These are the coordinates used in the completion engines and in Section 7.

Theorem 2.1 (Theorems A and B of the earlier release [`dep:THM-AB`])**.**

1. *For every positive definite integral M, $\theta_{L_0} = \theta_{L_1} = \theta_{L_2}$.*
2. *For every generic M, the lattices L_0, L_1, L_2 are pairwise non-isometric.*

Their proofs in the earlier release combine symbolic identities with finite exact checks. The earlier release's verifier rechecks all of them, and the release manifest reruns that verifier [`dep:THM-AB`].

Normalized coordinates and auxiliary forms. For generic seeds put $\beta = b/a \in (0, \frac{1}{2})$ and $\gamma = c/a > 1$. We use the linear forms

$$e_1 = a + c - b, \quad e_2 = a + c + b, \quad e_3 = 3a + c - 3b, \quad e_4 = a + 3c - 3b.$$

For each graph j, R_j denotes the span of all vectors $x \in L_j$ with $Q(x) \le 4a$.

The main question is whether a theta mate K of L_0 must be isometric to some L_j. The chamber arguments for $D \equiv 2 \pmod 3$ recover the short-vector structure of K up to a fixed window, and then complete it to rank four.

3. Primitive seeds with $D \not\equiv 2 \pmod 3$

Write $A = A(M)$ for the *ambient* lattice (E^2, Q), whose Gram matrix is $M \otimes A_2$. Each L_j is a sublattice of A of index 9. The residue cases rest on recovering an ambient theta mate from a theta mate of L_0, and on the rigidity of A.

Theorem 3.1 (Global ambient rigidity, R1; computer-assisted [`dep:RES-R1`])**.** *For every generic integer seed M, every theta mate of $A(M)$ is isometric to $A(M)$.*

Theorem 3.2 ($D \equiv 1 \pmod 3$; congruence reduction; computer-assisted [`dep:RES-D1`])**.** *Let M be primitive and generic, with $D \equiv 1 \pmod 3$, and let K be a theta mate of L_0. Put*

$$S_K = \{v \in K : \ Q(v) \equiv 0 \pmod 3\}, \qquad A_K = \tfrac{1}{3} S_K.$$

(Here $\frac{1}{3}S_K$ divides vectors and keeps the quadratic form.) Then:

- *A_K is an even integral lattice with $\theta_{A_K} = \theta_A$;*
- *$3A_K \subset K \subset A_K$, with $[A_K : K] = 9$.*

By Theorem 3.1, $A_K \cong A$. So K is an index-nine plane preimage in A, and the within-seed classification of plane preimages identifies K with some L_j. That classification is an exact finite certificate: a coset-minimum table, seven explicit isometries, and the separation on eleven parameter planes.

Theorem 3.3 ($D \equiv 0, 3 \pmod 9$ [`dep:RES-D0`])**.** *Let M be primitive and generic, with $D \equiv 0$ or $3 \pmod 9$. Two visible descents recover A from any theta mate of L_0 (with a finite residue certificate), and every theta mate is isometric to some L_j.*

Theorem 3.4 ($D \equiv 6 \pmod 9$ [`dep:RES-D6`])**.** *Let M be primitive and generic, with $D \equiv 6 \pmod 9$.*

- *Exact-three reduces to the rigidity statement (R_Z) for a visible Eisenstein closure Z, through two uniform ascents.*
- *(R_Z) holds on all such seeds (computer-assisted).*

Hence every theta mate is isometric to some L_j.

Remark 3.5 (Independent finite evidence). These theorems do not depend on the following, which remain independent checks for them [`dep:REG`]. (For $D \equiv 2 \pmod 3$ the register is part of the proof; see Remark 4.2.)

- the bounded register: every generic integer seed with $a \le 10$ below the tail (above it, the tail theorem applies), and all $D \equiv 1$ seeds with $a \le 14$;
- the finite register checks.

Their proofs are long, and were accepted separately. The table of sources in Appendix A gives, for each theorem:

- the proof note;
- the review in which it was accepted;
- for the computer-assisted parts, the sealed data and the verifier.

4. Primitive seeds with $D \equiv 2 \pmod 3$: chambers and walls

Let $M = (a, b, c)$ be primitive with $0 < 2b < a < c$ and $D = ac - b^2 \equiv 2 \pmod 3$. Put

$$\beta = \frac{b}{a} \in \left(0, \tfrac{1}{2}\right), \qquad \gamma = \frac{c}{a} > 1.$$

The six regions in Table 1 cover $\gamma > 1$ for every $\beta \in (0, \frac{1}{2})$. Each internal wall belongs to the row that ends at it.

Region	Exact interval	Proof
C4	$1 < \gamma \le 1 + \beta$	complete $4a$-span recognition and the C4 completion (§5)
C3	$1 + \beta < \gamma \le 1 + 3\beta$	recognition, the hybrid exclusion, and the completions (§6, §7)
C2	$1 + 3\beta < \gamma \le 3 - \beta$	the symbolic completion proof (§8)
C1	$3 - \beta < \gamma \le 3 + \beta$	first excess, binary recognition, rank-three stage and completion (§9)
C0 below the tail	$3 + \beta < \gamma \le (81 + 30\beta)/20$	the same chain with larger windows (§9)
tail	$\gamma > (81 + 30\beta)/20$	the tail theorem (Theorem 4.3)

Table 1. The chamber decomposition for primitive seeds with $D \equiv 2 \pmod 3$.

Remark 4.1 (Walls). The wall $c = a + b$ belongs to C4. The tail boundary $20c = 81a + 30b$ belongs to the C0 computation. Inside C3, graph 1 has two first-excess cells, joined along their own wall $2c = 2a + 3b$. That wall is computed separately, and keeps fourth vectors v with $Q(v) = F_0$ (§7). For each $\beta \in (0, \frac{1}{2})$, every $\gamma > 1$ lies in exactly one row.

Remark 4.2 (Small seeds). Some chamber arguments are stated for $a > 10$, or leave finitely many isolated points with $a \leq 10$. These are the C2 recognition and completion, the C1/C0 chain and the C3 graph-1 theorem. Every such primitive seed with $D \equiv 2 \pmod 3$ has $a \leq 10$, lies below the tail, and is one of the 228 seeds closed by the certificate-checked finite register [`dep:REG`] (Section 11). Seeds with $D \not\equiv 2 \pmod 3$ are covered by Section 3, and common scalings by Section 10. For $D \equiv 2 \pmod 3$ the register is therefore part of the proof, not only independent evidence.

Theorem 4.3 (Tail)**.** *If $20c > 81a+30b$, every theta mate of L_0 is isometric to one of L_0, L_1, L_2. This holds for every generic integer seed, including seeds with a common content. [`dep:TAIL`]*

The proof is a ladder. The region $4c > 17a + 6b$ (the window 19/2) is an earlier theorem, accepted at review (its separate small-row censuses are not used here). Two further rungs, at the windows 37/4 and 91/10, extend it to $20c > 81a + 30b$; each combines a complete streamer recovery with an exact coset band. The release manifest re-executes every rung:

- the rungs at 37/4 and 91/10 [`dep:TAIL`];
- the windows 10 and 19/2, rerun in their archived work tree (restored unchanged) against its recorded outputs, together with the closure test at $(5, 1, 23)$ and the bundled module for the windows 57/5 and 54/5 [`dep:TAIL19`];
- the bundled certificate modules above the window 57/5: Theorems D and E, the uniform conditions, and the shell, strip, window and boundary rungs [`dep:LADDER`].

The pattern of the finite reductions. In the new chambers (C4, C3, C1, and C0 below the tail) the proof follows one pattern.

(1) The short span of a theta mate K is identified. Either the complete span of the vectors with $Q \leq 4a$ is recognized (C4, C3; in C3 graph 1 it is followed by a forced first-excess pair), or forced first-excess pairs and a recognized binary span are found (C1, C0).
(2) Saturation and rounding lemmas give a primitive rank-three span, and a fourth basis vector of bounded norm with bounded pairings.
(3) Every remaining basis then lies in a finite symbolic branch universe, parametrized over closed domains in (β, γ): the closed chamber, or, for C3 graph 1, its two closed cells and the wall.
(4) Each branch is either:
    - excluded by a necessary determinant or theta condition, proved by an exact certificate on its whole closed domain; or

- identified with an explicit integral basis of a known L_j; or
- reduced to finitely many rational points, which are settled by exact theta differences and exact isometry tests.

(5) No branch remains unresolved ($U = 0$).

Agreement of finitely many theta coefficients is used only to keep candidates. *Every positive identification is an exact isometry*, never a theta-prefix assertion. Section 11 records the engines, the certificates and what has been independently re-executed.

5. Chamber C4

Throughout this section (a, b, c) is a seed of the closed chamber

$$\text{C4}: \qquad 0 < 2b < a < c \le a + b, \tag{1}$$

that is $0 < \beta < \frac{1}{2}$ and $1 < \gamma \le 1 + \beta$. The wall $c = a + b$ belongs to C4; the edges $c = a$ and $\beta = \frac{1}{2}$ are excluded. The set $0 < \beta < \frac{1}{2}$, $1 < \gamma \le 1 + \beta$ is the *generic chamber*: it contains every seed and the wall, and no point of an excluded edge. The generic chambers of the later sections are defined in the same way. We use the coordinates of Section 2: a vector is $v = (x, y, u, w) \in \mathbb{Z}^4$, standing for $(z_1, z_2) = (x + y\omega,\ u + w\omega)$, and

$$Q(v) = a\,\mathrm{N}(x, y) + b\,T(v) + c\,\mathrm{N}(u, w),$$

$$\mathrm{N}(x, y) = x^2 + xy + y^2, \quad T(v) = 2xu + xw + yu + 2yw,$$

which is the form Q of Section 2, and $B(v, v') = Q(v + v') - Q(v) - Q(v')$. Gram matrices are those of B. The lattice L_j is cut out by $(u, w) \equiv \Phi_j(x, y) \pmod 3$.

5.1. **The short vectors.** On C4 the four values $e_1 < e_2 < e_3 < e_4$ are all at most $4a$: indeed $e_4 = a + 3c - 3b \le 4a$, with equality exactly on the wall. They are pairwise distinct on the generic chamber: $e_2 - e_1 = 2b$, $e_3 - e_2 = 2(a - 2b)$ and $e_4 - e_3 = 2(c - a)$ vanish only on the excluded edges.

Lemma 5.1 (The short vectors of L_j)**.** *At every point of* C4*, the vectors of L_j with $Q \le 4a$ are exactly $\pm g_1, \dots, \pm g_4$, with $Q(g_i) = e_i$, where*

graph	$g_1,\ g_2$	$g_3,\ g_4$	*relation*
0	$(1,0,0,-1),\ (0,1,1,0)$	$(2,-1,-1,1),\ (1,-1,-1,2)$	$g_4 = g_3 - g_1$
1	$(1,-1,0,1),\ (0,1,1,0)$	$(1,-2,-1,1),\ (1,0,-2,1)$	$g_3 = g_1 - g_2$
2	$(1,0,-1,1),\ (0,1,1,0)$	$(1,-2,0,1),\ (1,-1,-2,1)$	$g_4 = g_1 - g_2$

So R_j has rank three. A basis of R_j is (g_1, g_2, g_3) for graphs 0 *and* 2*, and (g_1, g_2, g_4) for graph* 1*. In each case its Gram matrix H_j has* $\det H_j = 54De_1$.

Proof. The norms, the congruences and the relations are direct substitutions. The determinant is a direct expansion; the off-diagonal entries of H_j are $(a - c,\ 3a - b - c,\ 4b - c)$, $(-a + 3b + c,\ a - 2b,\ a + 2b - 3c)$ and $(a + 3b - c,\ c - 2b,\ -3a + 2b + c)$ for (B_{12}, B_{13}, B_{23}) in graphs $0, 1, 2$, with the basis vectors listed above. That no other vector has $Q \le 4a$ follows from

the complete catalogue of L_j below $12a$ (the finite box of all vectors whose norm is below $12a$ somewhere in the generic region $0 < 2b < a < c$), by comparing each catalogue form with $4a$ at the vertices of the closed chamber [dep:C4-REC]. □

A *theta mate* of L_0 is a positive definite even lattice K of rank four with $\theta_K = \theta_{L_0}$; since $\theta_{L_0} = \theta_{L_1} = \theta_{L_2}$ we write θ_L. Then $\det K = (27D)^2$, since the theta series determines the rank and the determinant. By Lemma 5.1 and the distinctness of the e_i, a mate K has exactly eight vectors $\pm g'_1, \dots, \pm g'_4$ with $Q \le 4a$, and $Q(g'_i) = e_i$. Its *complete* $4a$ *span* is $R = \sum_i \mathbb{Z} g'_i$.

5.2. **Recognition of the complete** $4a$ **span.**

Theorem 5.2 (Recognition in C4)**.** *Let (a, b, c) be a primitive seed of the closed chamber* (1) *with $D \equiv 2 \pmod 3$, and K a theta mate of L_0. Then the complete $4a$ span R of K is isometric to R_j for some $j \in \{0, 1, 2\}$. In particular R has rank three and satisfies the triangle relation of Lemma 5.1 for that j. No span of rank four occurs. [dep:C4-REC]*

The proof is computer-assisted. It has three steps.

Step 1: pair control. The largest pair sum on the closed chamber is $e_3 + e_4 = (4 - 6\beta + 4\gamma)a \le (8 - 2\beta)a < 8a$. Hence $Q(g'_i \mp g'_j) = e_i + e_j - |B(g'_i, g'_j)| < 8a$ for the appropriate sign, and this is a nonzero value of θ_L below $9a$. So each of the six pairings has the form $B_{ij} = \pm(e_i + e_j - F_{ij})$, where F_{ij} is one of the finitely many affine forms of the catalogue. A choice of the six forms and signs is a *span branch*.

Step 2: the exact symbolic classification. Each span branch is tested on an exact closed convex polygon, segment or point in the (β, γ)-plane, in rational arithmetic. For every integer vector x in a box, the norm $Q(\sum_i x_i g'_i)$ is an integer affine form ℓ_x, and the following necessary conditions are imposed.

- (C0) A relation x satisfies $Gx = 0$ for the 4×4 Gram matrix G of (g'_i).
- (C1) ℓ_x is 0, a value of θ_L below the window, or at least the window.
- (C2) $0 < \ell_x \le 4a$ forces $\ell_x = e_k$ and $x \mp u_k$ to be a relation, u_k the k-th unit vector.
- (C3) At each value below the window, the number of vectors of the span is at most the coefficient of θ_L.

A condition that holds only where two forms agree restricts the domain to that line or point, so coincidence lines are exact, not sampled. Every exclusion is valid at every point of its domain, so the survivors form a *superset* of all possible spans. The run used a box of radius 2 with window $9a$, and then a box of radius 3 with window $12a$ on the survivors. Its outcome:

- every survivor on a two-dimensional domain is identically one of the known Grams H_0, H_1, H_2, up to signs;
- the other survivors lie on 36 isolated points, at ten seeds, and on no segment and no region;

- no survivor of rank four remains at any stage. The determinant $(27D)^2$ is not used in this step.

Every known span survives on the whole closed chamber.

Step 3: the isolated seeds. The ten seeds are $(3,1,4)$, $(5,1,6)$, $(5,2,6)$, $(6,2,7)$ and $(12,2,13)$, with $D \equiv 2 \pmod 3$, and five seeds with $D \not\equiv 2 \pmod 3$, which are not needed (Section 3). At each of the five in-scope seeds the alternative spans were enumerated exactly and completed.

- At $(12,2,13)$ the predicted alternative fails the exact span conditions, so no alternative exists.
- At the other four, each alternative span either has no even integral completion of determinant $(27D)^2$, or every such completion differs from θ_L below $9a$. As positive controls, every known span has completions that agree with θ_L through $9a$.

This proves Theorem 5.2. The seeds with $a \le 10$ among these are also closed by the register (Remark 4.2).

Remark 5.3 (Scope of the recognition). (1) The classification uses the necessary target counts of θ_L *through* $12a$, after a first stage at $9a$. It is not a consequence of the $9a$ prefix alone.

(2) The counts of the span itself do not recognize it. An exact enumeration at all 34 primitive seeds of C4 with $a \le 10$ finds span Grams other than the known ones, with no excess over θ_L through $12a$, at 7 seeds (4 with $D \equiv 2 \pmod 3$). The completion step is what removes them.

(3) The triangle relation of an arbitrary mate is a consequence of the classification, not of frame algebra alone.

(4) The same exact enumeration is a cross-validation of Step 2: at those 34 seeds the symbolic record predicts exactly the spans found (sound and sharp at 34/34).

The following elementary fact explains why a 3-frame can miss the complete span.

Lemma 5.4 (Frames). *Let $R = \sum_{i=1}^{4} \mathbb{Z} g_i$ have rank three with primitive relation $x \in \mathbb{Z}^4$, and let $F_l = \sum_{i \neq l} \mathbb{Z} g_i$. Then $R/F_l \cong \mathbb{Z}/x_l\mathbb{Z}$. So $F_l = R$ if and only if $|x_l| = 1$, and F_l has rank two if $x_l = 0$.*

Proof. The kernel of $\mathbb{Z}^4 \to R$ is saturated of rank one, hence equal to $\mathbb{Z}x$. So $R \cong \mathbb{Z}^4/\mathbb{Z}x$, and the image of $\bigoplus_{i \neq l} \mathbb{Z} u_i$ has quotient $\mathbb{Z}/x_l\mathbb{Z}$. □

With the triangle relations, three of the four frames are bases of R, and the fourth has rank two.

5.3. **The fourth basis vector.**

Lemma 5.5 (Rounding). *Let $s_1, \dots, s_k$ be a basis of a lattice S and $y \in \mathbb{R}S$. Then $y + S$ contains a vector z with $Q(z) \le \frac{1}{4} \sum_i Q(s_i)$.*

Proof. Nearest-plane rounding along the Gram–Schmidt vectors $s_k^*, \dots, s_1^*$ gives $z = \sum_i c_i s_i^*$ with $|c_i| \le \frac{1}{2}$. The s_i^* are orthogonal and $Q(s_i^*) \le Q(s_i)$. □

Lemma 5.6 (Saturation)**.** *Let K be a lattice and R the span of the vectors of K with $Q \le Y$. Suppose R has a basis $s_1, \dots, s_k$ with $k \le 3$ and $Q(s_i) \le Y$. Then R is primitive in K, that is $K \cap \mathbb{R}R = R$.*

Proof. Otherwise Lemma 5.5 gives $z \in (K \cap \mathbb{R}R) \setminus R$ with $Q(z) \le kY/4 < Y$, so $z \in R$ by the definition of R, a contradiction. □

Lemma 5.7. *Let R be the complete Y-span of K and $K = R \oplus \mathbb{Z}v$. Then $Q(x + kv) > Y$ for every $x \in R$ and every $k \ne 0$.*

Proof. $x + kv \notin R$, and R contains every vector of K with $Q \le Y$. □

By Theorem 5.2, $R \cong R_j$ has a basis s_1, s_2, s_3 of vectors with $Q \le 4a$: (g_1', g_2', g_3') for graphs $0, 2$ and (g_1', g_2', g_4') for graph 1. By Lemma 5.6, R is primitive in K, so $K = R \oplus \mathbb{Z}v$ for any v whose class generates $K/R \cong \mathbb{Z}$.

Proposition 5.8 (The fourth basis vector [`dep:BASIS-BOUNDS`])**.** *Let $R \cong R_j$ be the complete $4a$ span of a theta mate K, at a point of the closed chamber* C3 ∪ C4 *(that is $a < c \le a + 3b$), with $j \in \{0, 2\}$, or with $j = 1$ in* C4*. Choose v shortest in its coset $v + R$, where $v + R$ generates K/R. Put $f = Q(v)$, $u_i = B(s_i, v)$ and $t_i = Q(s_i) + f - |u_i|$. Then*

$$4a < f < \tfrac{125}{16}\, a < 8a, \qquad 4a < f \le t_i \le f + Q(s_i) < 12a,$$

and (s_1, s_2, s_3, v) is a basis of K, so that

$$\det \begin{pmatrix} H_j & u \\ u^{\mathsf{T}} & 2f \end{pmatrix} = (27D)^2. \tag{2}$$

Proof. Write $v = y + z$ with $y \in \mathbb{R}R$ and $z \perp R$. Then $B(z, z) = \det K / \det R = (27D)^2/(54De_1) = 27D/(2e_1)$, and Lemma 5.5 lets us replace v by a representative of the same coset with $Q(v) \le \frac{27D}{4e_1} + \frac{1}{4}\sum_i Q(s_i)$. For $c \le a + 3b$,

$$3ae_1 - 4D = a(3a - 3b - c) + 4b^2 \;\ge\; 2(a - b)(a - 2b) > 0,$$

so $\frac{27D}{4e_1} < \frac{81}{16}a$. The norm sum is $e_1 + e_2 + e_3 = 5a - 3b + 3c < 11a$ for graphs $0, 2$, and $e_1 + e_2 + e_4 = 3a - 3b + 5c < 9a$ for graph 1 in C4. Hence $f < \frac{81}{16}a + \frac{11}{4}a = \frac{125}{16}a$ for a shortest representative. The lower bound $f > 4a$ is Lemma 5.7 with $k = 1$, $x = 0$. Minimality of v in $v + R$ gives $Q(v \pm s_i) \ge Q(v)$, that is $|u_i| \le Q(s_i)$, and $t_i = \min_{\pm} Q(v \pm s_i)$; Lemma 5.7 gives $t_i > 4a$, and $t_i \le f + Q(s_i) < \frac{125}{16}a + 4a < 12a$. Finally R is primitive of rank three in K of rank four, so (s_1, s_2, s_3, v) is a basis and (2) is $\det K$. □

5.4. **The completion universe.** Proposition 5.8 reduces a theta mate to finitely many symbolic data. Since f and the t_i are values of θ_L below $12a$, each is the value of an affine form of the catalogue. A *branch* is a choice of:

- the graph j of R;

- catalogue forms f, t_1, t_2, t_3;
- signs $\sigma = (\sigma_1, \sigma_2, \sigma_3)$ with $u_i = \sigma_i(Q(s_i) + f - t_i)$, modulo $v \mapsto -v$, so $\sigma_1 = +1$.

Its *domain* is the closure, within the closed chamber, of the set where $4a < f < 8a$, $f \le t_i \le f + Q(s_i)$ and $4a < t_i < 12a$: an exact polygon, segment or point. Taking closures only enlarges the domains. On the domain, $G(\beta, \gamma) = \left(\begin{smallmatrix} H_j & u \\ u^{\mathsf{T}} & 2f \end{smallmatrix}\right)$ has affine entries, and $P = \det G - (27D)^2$ is a polynomial of degree four in (β, γ) after dividing by a^4. A mate gives a rational point of some branch domain at which $P = 0$ and all further necessary conditions hold. The branches with the same (j, f, t_1, t_2) form a *unit*.

How a branch is decided. Each branch receives one or more of the following outcomes, each an exact statement about its closed domain (for R, O, L and P, about its rational points in the generic chamber); a branch with several pieces or points may receive several.

- **N:** P has no zero on the closed domain. The certificate subdivides the box $\beta \in [0, \frac{1}{2}]$, $\gamma \in [1, 3]$ dyadically, to depth at most 6. Boxes outside the domain's half-planes are discarded. On each remaining box an exact enclosure of P, valid because every monomial is monotone for $\beta, \gamma \ge 0$, shows that P has a constant strict sign. The box contains the closed chamber; see Section 11 for the domain guard.
- **R:** no eligible rational point remains: for $P \not\equiv 0$ the line cuts below leave none, and for an identity family ($P \equiv 0$ on the domain) the resolver excludes every piece. A test vector $x = x_1 s_1 + x_2 s_2 + x_3 s_3 + v$ has a norm N_x, an affine form. Where $N_x < 16a$, a mate needs N_x to be one of the finitely many target forms, so the branch lives on the lines $N_x = \tau$. On each such segment P restricts to a univariate polynomial, and its rational roots are the only candidates. The part where $N_x \ge 16a$ passes to the next test vector. An R decision concerns the eligible generic points of the chamber (Remark 11.1).
- **O:** rational points exist only outside the generic chamber, that is on excluded edges or at vertices.
- **L:** rational points exist, and each fails the value test: one of the 27 test vectors x with $x_i \in \{-1, 0, 1\}$ has $N_x < 16a$ not a value of θ_L.
- **P:** candidate points, settled by exact theta comparison and an exact isometry test.
- **K:** an identity family, $P \equiv 0$ on a piece of the domain, identified with an explicit integral basis of a known L_j by an identity of forms.

A branch on which none of these applies is *unresolved* (U). Two details matter for completeness.

- *Satisfied equalities are kept.* A restriction that holds identically on the current segment or point is retained, not skipped; 1,538,500 such events were processed.

- *Families are decided by their resolvers.* If $P \equiv 0$ on the domain, or on a cut line, a two-dimensional or one-dimensional resolver applies the value test, the short-shell rule and the count rule to the family. A surviving piece is identified (K), is a single point (P), or is saved (U). A family point excluded by the count rule is not listed as a point record, even if it passes the 27-vector value test.

	N	R	O	L	P	K	U
branches	5,677,929	900,764	31,543	1,139	385	192	0

TABLE 2. The C4 completion: 58,692 units and 6,611,512 branches (2,177,996, 2,255,520 and 2,177,996 for graphs 0, 1, 2). The P branches carry 743 point records; the K branches carry 308 identified pieces. A branch can carry several letters.

Theorem 5.9 (C4)**.** *Let (a, b, c) be a primitive seed of the closed chamber* (1) *with $D \equiv 2 \pmod 3$, the wall $c = a + b$ included. Then every theta mate of L_0 is isometric to L_0, L_1 or L_2. More precisely, a mate whose complete $4a$ span is R_j is isometric to L_j. [dep:C4]*

Proof. By Theorem 5.2, the span of a mate K is $R \cong R_j$, and by Lemma 5.6 and Proposition 5.8, K has a basis (s_1, s_2, s_3, v) whose Gram matrix is the matrix G of some branch, evaluated at the seed. The seed then lies in that branch's domain and satisfies every necessary condition used there. The completion run of Table 2 decides every branch, with $U = 0$ [dep:UNIVERSES].

- An N, R, O or L branch, or a family piece excluded by its resolver, contains no such seed.
- A K piece is L_j in an explicit integral basis. At points of the generic chamber every K piece is identified with the L_j of its own span.
- The 743 point records give 233 distinct pairs (graph, Gram) at their seeds, 76 of them with $D \equiv 2 \pmod 3$, each with $\det G = (27D)^2$. Of these, 219 agree with θ_L through $16a$, and each of the 219 is isometric, by an exact isometry test, to the L_j of its own span (98, 78 and 43 for $j = 0, 1, 2$, at seeds up to $a = 78$). The other 14 differ from θ_L first between $4.2a$ and $4.6a$; all of them have $a \le 10$.

So every mate is isometric to some L_j. □

5.5. **Controls and the independent diagnostic.** The run carries the following predeclared controls, all of which pass.

- **K1** (L_j is found). At each of 153,010 seeds, for each graph, a K piece of that graph contains the seed. The seeds are the 152,994 of the bounded census of C4 (all primitive seeds with $D \equiv 2 \pmod 3$ in the closed chamber with $a \le 200$, and those on the wall with

$201 \le a \le 600$), every primitive seed of the closed chamber with $a \le 10$, and the control seeds.

- **K2** (the branch universe). At 30 pairs (control seed, graph), a brute-force search for integral completions, by a different method, gives exactly the recorded symbolic hits.
- **K2′** (end to end). Every brute-force completion whose theta series agrees with θ_L through $16a$ is found by the engine, as a K piece containing the seed or as a point record: 30 of 30.
- **K3** (the register and the bounded census). None of the 107 candidates at seeds with $a \le 10$, or in the bounded census, contradicts either of them [`dep:REG`], [`dep:C4-200`].
- **Edges.** The 45 distinct O points lie at the vertices $(0,1)$, $(\frac{1}{2},1)$, $(\frac{1}{2},\frac{3}{2})$, on the excluded edge $c=a$ and on $\beta=\frac{1}{2}$. None lies in the generic chamber; in particular the wall $c=a+b$ never carries an O decision.
- **Identity events.** On a pilot of 70 units the restriction contract was checked directly: identity events occur on points and segments only, never on two-dimensional pieces, and they are all processed by the line cuts.

Remark 5.10 (The independent diagnostic). An independent review found 13 rational points, six in C4 (at $(7,3,9)$) and seven in C3, that pass its value test inside a branch but carry no point record. All 13 lie in identity families or on one-dimensional families, where the resolver, not the point test, decides. Each was excluded by the resolver's count rule, and each has an explicit *count witness*: a value $n < 16a$ carrying, within the resolver's box, more pairs of test vectors than $\frac{1}{2}r_L(n)$, where $r_L(n)$ is the coefficient of q^n in θ_L. Since the four basis vectors form a basis, distinct test vectors modulo sign are distinct lattice vectors, so this is a valid necessary condition. The witnesses were recomputed and checked independently [`dep:C4`].

Remark 5.11 (The excluded edge). A general re-identification of all 308 known pieces, through Gram matrices only, confirmed every one of them. On the excluded edge $c=a$, 30 pieces of graph-2 spans were identified with an explicit basis of L_1 by the general route, and with L_2 by the span's own vectors; both are integral isometries there. The edge carries no seed.

Remark 5.12 (Relation to the bounded census). The bounded census of C4 (152,994 seeds) is a separate bounded theorem. It shares the recognition step (Theorem 5.2) and the saturation lemma, and replaces the completion search by the finite separation of multiplier completions [`dep:C4-200`]. Theorem 5.9 supersedes it inside C4; the census remains a check of the completion step only. Theorem 5.9 does not prove a uniform first-difference bound for multiplier completions: where such a completion exists, the search shows only that it is not a theta mate.

Section 11 records the engine and what has been re-executed independently [dep:ENG].

6. Chamber C3: recognition, the hybrid exclusion, graphs 0 and 2

Throughout this section (a, b, c) is a seed of the closed chamber

$$\text{C3}: \qquad 0 < 2b < a, \qquad a + b < c \le a + 3b, \tag{3}$$

that is $0 < \beta < \frac{1}{2}$ and $1 + \beta < \gamma \le 1 + 3\beta$. The wall $c = a + 3b$ belongs to C3, and the edge $c = a + b$ belongs to C4. Coordinates, Q, B and the vectors g_i of each L_j are those of Section 5.

The short vectors. On C3 we have $e_1 < e_2 < e_3 \le 4a < e_4$: the inequality $e_3 \le 4a$ is $c \le a + 3b$, with equality exactly on the wall, and $e_4 > 4a$ is $c > a + b$. The vectors of L_j with $Q \le 4a$ are exactly $\pm g_1, \pm g_2, \pm g_3$ (Lemma 5.1 and the catalogue). So:

- for graphs 0 and 2, $R_j = \mathbb{Z}g_1 + \mathbb{Z}g_2 + \mathbb{Z}g_3$ has rank three and Gram matrix H_j, with $\det H_j = 54De_1$;
- for graph 1, $g_3 = g_1 - g_2$, so $R_1 = \mathbb{Z}g_1 + \mathbb{Z}g_2$ is binary.

A theta mate K has exactly six vectors $\pm g'_1, \pm g'_2, \pm g'_3$ with $Q \le 4a$, $Q(g'_i) = e_i$, and its complete $4a$ span is $R = \mathbb{Z}g'_1 + \mathbb{Z}g'_2 + \mathbb{Z}g'_3$.

6.1. **Recognition.**

Theorem 6.1 (Recognition in C3)**.** *Let (a, b, c) be a primitive seed of the closed chamber* (3) *with $D \equiv 2 \pmod 3$, and K a theta mate of L_0. Then the complete $4a$ span of K is isometric to R_0, R_1 or R_2. [dep:C3-REC-HYB]*

The proof follows that of Theorem 5.2.

- *Pair control.* The largest pair sum on the closed chamber is $e_2 + e_3 = (4 - 2\beta + 2\gamma)a \le (6 + 4\beta)a < 8a$. So each pairing is $B_{ij} = \pm(e_i + e_j - F_{ij})$ for a catalogue form F_{ij}.
- *The symbolic classification*, with the conditions (C0)–(C3) of Section 5.2, at window $9a$ and then $12a$. Every survivor on a two-dimensional domain is a known span, and every known span survives on the whole closed chamber. The other survivors lie on 75 isolated points, at 29 seeds, and on exactly two segments.
- *The isolated seeds* with $D \equiv 2 \pmod 3$ are $(3, 1, 5)$, $(3, 1, 6)$, $(6, 1, 8)$, $(7, 1, 9)$, $(7, 2, 12)$, $(7, 3, 11)$, $(11, 3, 16)$, $(11, 5, 21)$ and $(20, 8, 39)$. At each, the alternatives were enumerated exactly and completed. Each alternative has no even integral completion of determinant $(27D)^2$, or all its completions differ from θ_L below $9a$. Of the other isolated seeds, $(9, 1, 10)$ lies on the edge $c = a + b$, which belongs to C4, and the rest have $D \not\equiv 2 \pmod 3$ (Section 3).
- *The two segments* carry one hybrid span (Lemma 6.2), which is excluded by the hybrid-exclusion theorem below (Theorem 6.3).

Cross-validation. An exact enumeration of spans at all 68 primitive seeds of the closed chamber with $a \le 10$, and at $(11, 5, 21)$, lies inside the symbolic prediction at every seed (sound at 69/69). As in C4, the counts of the span itself do not recognize it; the completion steps do. The classification uses target counts through $12a$ (Remark 5.3).

Lemma 6.2 (The hybrid span)**.** *The two segments of the classification are*

$$c = 4b, \quad \tfrac{5}{12} \le \beta < \tfrac{1}{2}, \qquad \text{and} \qquad c = 3a - 2b, \quad \tfrac{2}{5} \le \beta < \tfrac{1}{2}.$$

On both, the surviving span has the single Gram matrix

$$H = \begin{pmatrix} 2e_1 & c-a & c-2b \\ c-a & 2e_2 & 3a+b-c \\ c-2b & 3a+b-c & 2e_3 \end{pmatrix}.$$

Its determinant is $\det H = 54b(4a^2+9ab-b^2)$ *on* $c = 4b$ *and* $\det H = 54(a-b)(12a^2-7ab-b^2)$ *on* $c = 3a-2b$*, so* $\det H - 54De_1$ *equals* $-108b^2(a-b)$*, respectively* $-108b(a-b)^2$*. Hence* H *is isometric to no known span there. On both lines* $D \equiv b(a-b) \pmod 3$*, so* $D \equiv 2 \pmod 3$ *exactly when* $3 \mid a$ *and* $3 \nmid b$*.*

Proof. The segments and the assignment are the output of the classification [`dep:C3-REC-HYB`]. The determinants are direct expansions. On $c = 4b$, $D = b(4a-b)$; on $c = 3a-2b$, $D = (a-b)(3a+b)$; in both cases $D \equiv b(a-b) \pmod 3$, and $b(a-b) \equiv 2$ forces $a \equiv 0$, $b \not\equiv 0 \pmod 3$. □

The entry $B_{12} = c - a$ is graph 0's, up to sign, and $B_{13} = c - 2b$ is graph 2's, which is why we call H a hybrid.

6.2. **The hybrid-exclusion theorem.**

Theorem 6.3 (Hybrid exclusion)**.** *Let* (a, b, c) *be a primitive seed with* $D \equiv 2 \pmod 3$ *on one of the segments of Lemma 6.2. Then no theta mate of* L_0 *has the hybrid span* H *as its complete* $4a$ *span. [`dep:C3-REC-HYB`]*

Lemma 6.4 (A short fourth vector)**.** *Let* K *be an even positive definite lattice of rank four and determinant* $\Delta = (27D)^2$*, whose complete* $4a$ *span* R *has Gram matrix* H*, on one of the two segments. Then* $K = R \oplus \mathbb{Z}v$ *with*

$$4a < Q(v) \le \frac{\Delta}{2 \det H} + \frac{e_1 + e_2 + e_3}{4} < 8a.$$

Proof. R is primitive (Lemma 5.6). Write a generator of K/R as $y + z$, with $y \in \mathbb{R}R$ and $z \perp R$; then $B(z, z) = \Delta / \det H$, and rounding y (Lemma 5.5) gives a generator v with the displayed upper bound. The strict inequality with $8a$ is $P := (32a - e_1 - e_2 - e_3) \det H - 2\Delta > 0$. On $c = 4b$, $P = 486b\,(12a^3 - 25a^2b + 12ab^2 - 2b^3)$, and $12 - 25\beta + 12\beta^2 - 2\beta^3$ decreases on $[\frac{5}{12}, \frac{1}{2}]$ to its minimum $\frac{9}{4}$. On $c = 3a - 2b$, $P = 486(a-b)(-3a^3 + 7a^2b + 6ab^2 + 2b^3)$, and $-3 + 7\beta + 6\beta^2 + 2\beta^3$ increases for $\beta \ge \frac{2}{5}$, with minimum $\frac{111}{125}$. Finally $Q(v) > 4a$ by Lemma 5.7. □

Proof of Theorem 6.3. Let K be such a mate, with basis (g'_1, g'_2, g'_3, v) as in Lemma 6.4, $f = Q(v)$ and $u_i = B(g'_i, v)$. Then $t_i = \min_{\pm} Q(g'_i \pm v) = e_i + f - |u_i|$ satisfies $t_i > 4a$ (Lemma 5.7) and $t_i \le e_i + f < 12a$, because $e_i \le 4a$ on the closed segments. So f and the t_i are values of θ_L below $12a$.

- *Branches.* On each line, each catalogue form of L_0 below $12a$ restricts to a linear form in β; identical restrictions are merged, with their multiplicities added. A branch chooses such forms f, t_1, t_2, t_3 and signs σ_2, σ_3, with $\sigma_1 = +1$ modulo $v \mapsto -v$. Its domain is the exact subinterval of the segment where $4a < f < 8a$, $4a < t_i < 12a$ and $t_i \le e_i + f$. The ends $\beta = \frac{5}{12}$ and $\beta = \frac{2}{5}$ are closed, and $\beta = \frac{1}{2}$ is open.
- *Exhaustiveness.* At the seed, the true values $Q(v)$ and $Q(g'_i \pm v)$ are target values, so the branch carrying these forms contains the seed, and $\det G = \Delta$ holds there, with $G = \left(\begin{smallmatrix} H & u \\ u^{\mathsf{T}} & 2f \end{smallmatrix}\right)$. So either $\det G - \Delta$ vanishes identically on the branch, or $\beta = b/a$ is one of its rational roots in the domain.
- *The search.* There are 687,500 branches on $c = 4b$ and 296,352 on $c = 3a - 2b$; 249,224 have a nonempty domain. On every one of them $\det G - \Delta$ is a nonzero polynomial: there is no identity branch. Only 19 and 70 branches have a rational root in their domain, at seven primitive seeds on each line. Among these, only $(9, 4, 16)$ and $(105, 46, 223)$ have $D \equiv 2 \pmod 3$.
- *The candidates.* Every candidate Gram matrix at those seeds has determinant Δ, and differs from θ_L below $9a$: at $(9, 4, 16)$ first at 45, 51, 51 and 59 against $9a = 81$, and at $(105, 46, 223)$ at 636 against 945. Over all 37 candidates, in scope or not, the largest first difference is $7.03a$.

So no such mate exists. □

Controls. The same search applied to the known spans H_0 and H_2 on both lines finds identity families, identified with explicit integral bases of L_0 and L_2, whose domains cover each segment. A brute-force enumeration of completions, by a different method, at all 13 primitive seeds with $a \le 15$ on the segments gives the same $(f, \pm u)$ sets as the certificate, for all three spans (39 of 39). Both endpoints are covered, and the brute force finds no candidate at them. The producer run was reproduced in full by a reviewer. An independent audit checked the branch universe. A second independent audit covered every branch exactly once, reconstructed the determinant coefficients, checked every rational root, enumerated all 37 exclusions anew and verified 42 positive-control isometries.

6.3. **Graphs** 0 **and** 2**: Theorem** C3(0, 2)**.**

Theorem 6.5 (C3(0, 2))**.** *Let (a, b, c) be a primitive seed of the closed chamber* (3) *with $D \equiv 2 \pmod 3$, the wall $c = a + 3b$ included. Let K be a theta*

mate of L_0 whose complete $4a$ span is isometric to R_0 or R_2. Then K is isometric to L_0, L_1 or L_2; in fact $K \cong L_j$ when its span is R_j. [dep:C3-02]

Proof. The span $R \cong R_j$ has the basis (g_1', g_2', g_3') of vectors with $Q \le 4a$, so it is primitive (Lemma 5.6). Proposition 5.8, which holds for $c \le a+3b$, gives $K = R \oplus \mathbb{Z}v$ with $4a < f < \frac{125}{16}a$ and $4a < t_i < 12a$. The branch universe is that of Section 5.4, on the closed chamber (3), for graphs 0 and 2, and every branch is decided in the same way (Table 3), with $U = 0$ [dep:UNIVERSES].

- Every K piece is the L_j of its own span, in an explicit integral basis.
- The 932 point records give 214 distinct (graph, seed, Gram), 67 with $D \equiv 2 \pmod 3$, each with $\det G = (27D)^2$. Of these, 198 agree with θ_L through $16a$, and each is isometric, by an exact isometry test, to the L_j of its own span (108 copies of L_0 and 90 of L_2, at seeds up to $a = 47$). The other 16 differ from θ_L first between $4.67a$ and $7.0a$, at seeds with $a \le 7$.

□

	N	R	O	L	P	K	U
branches	3,056,679	154,660	10,764	1,191	448	93	0

TABLE 3. The C3 completion for graphs 0 and 2: 27,844 units and 3,223,584 branches (1,611,792 per graph). The P branches carry 932 point records, and the K branches carry 99 identified pieces; 212,703 identically satisfied restrictions were kept and processed. A branch can carry several letters.

Controls, all predeclared and all passing:

- **K1**: at 72 seeds, namely the 68 primitive seeds of the closed chamber with $a \le 10$ and four control seeds with $a > 10$, for graphs 0 and 2, a K piece of that graph contains the seed. A larger positive control repeats this at 267,133 primitive seeds with $D \equiv 2 \pmod 3$: those with $a \le 200$, and those on the wall $c = a + 3b$ with $201 \le a \le 600$. It shows that L_j is found; it is not a coverage input.
- **K2**: brute force equals the symbolic hits at 20 of 20 pairs (control seed, graph).
- **K2′**: every brute-force completion matching θ_L through $16a$ is found (21 of 21).
- **K3**: none of the 81 candidates with $a \le 10$ contradicts the register [dep:REG].
- **Edges**: the 24 O points lie at the vertices $(0, 1)$, $(\frac{1}{2}, \frac{3}{2})$, $(\frac{1}{2}, \frac{5}{2})$, on the excluded edges $\beta = \frac{1}{2}$ and $c = a + b$. None lies in the generic chamber; in particular the wall $c = a+3b$ never carries an O decision.

Graph 1. A mate whose span is isometric to the binary R_1 is treated in Section 7. Together with Theorem 6.5, that section completes the exact-three statement on the closed chamber.

7. Chamber C3, graph 1

This section treats the theta mates in the chamber C3 whose short span is binary. Throughout, (a,b,c) is a seed of the closed chamber

$$\text{C3}: \qquad 0<2b<a, \qquad a+b<c\le a+3b, \tag{4}$$

with $\beta=b/a$, $\gamma=c/a$ and $D=ac-b^2$. The edge $c=a+b$ belongs to C4, and the edge $c=a+3b$ belongs to C3.

7.1. **Coordinates, the binary span and the statement.** For $X=(x,y)$ and $U=(u,w)$ in $\mathbb{Z}^2$ put

$$\mathrm{N}(X)=x^2+xy+y^2, \qquad T(X,U)=2xu+xw+yu+2yw,$$

so that $\mathrm{N}(X+U)=\mathrm{N}(X)+\mathrm{N}(U)+T(X,U)$. The form N takes the values $0,1,3,4,7,9,\dots$ and never 2, 5 or 6. We work with L_1 in the ambient coordinates

$$L_1=\{(X,U)\in\mathbb{Z}^4:\ u\equiv x+y,\ w\equiv x \pmod 3\},$$
$$Q(X,U)=a\,\mathrm{N}(X)+b\,T(X,U)+c\,\mathrm{N}(U).$$

Its bilinear form is $B(v,v')=Q(v+v')-Q(v)-Q(v')$. Gram matrices G are taken with respect to B, so their diagonal entries are $2Q$, and $\det G_{L_1}=(27D)^2$. Theta series count vectors by Q: $\theta_L=\sum_{v\in L}q^{Q(v)}$. The congruence reads $U\equiv MX \pmod 3$ with $M=\left(\begin{smallmatrix}1&1\\1&0\end{smallmatrix}\right)$. These are the coordinates of Section 2: $z_1=x+y\omega$, $z_2=u+w\omega$, so that $\mathrm{N}(X)=N(z_1)$, $T(X,U)=\mathrm{Tr}(\bar z_1 z_2)$ and the congruence is $z_2\equiv\Phi_1 z_1 \pmod 3$.

Put

$$g_1=(1,-1,0,1), \qquad g_2=(0,1,1,0),$$
$$v_4=(1,0,-2,1), \qquad h=(1,1,-1,1).$$

Then

$$e_1:=Q(g_1)=a-b+c, \qquad e_2:=Q(g_2)=a+b+c,$$
$$B(g_1,g_2)=-a+3b+c,$$

$Q(v_4)=a-3b+3c$, $\quad Q(h)=3a+c$, $\quad F_0:=\min(a-3b+3c,\ 3a+c)$. The plane $\Pi=\mathbb{R}g_1+\mathbb{R}g_2$ is $\{w=x,\ u=x+y\}$. Its integral points are $(x,y,x+y,x)=xg_1+(x+y)g_2$, so $L_1\cap\Pi=\mathbb{Z}g_1+\mathbb{Z}g_2$.

Cells and the tie wall. On the wall $2c=2a+3b$ one has $Q(v_4)=Q(h)$. It divides the closed chamber into the *lower cell* $2c<2a+3b$, where $F_0=Q(v_4)$, and the *upper cell* $2c>2a+3b$ (with $c\le a+3b$), where $F_0=Q(h)$. In the (β,γ)-plane the closures of the cells are the triangles with vertices $(0,1),(\frac12,\frac32),(\frac12,\frac74)$ and $(0,1),(\frac12,\frac74),(\frac12,\frac52)$. Every vertex lies outside C3.

Lemma 7.1. *On* C3*:*

(a) $0<B(g_1,g_2)<e_1<e_2<4a<F_0<\frac{11}{2}a$*;*

(b) $e_2 + F_0 < \frac{19}{2}a$;
(c) $a - b > b > 0$, $c - b > 0$ *and* $4(a - b) > 3b$.

Proof. For (a):

- $B(g_1, g_2) > -a+3b+(a+b) = 4b > 0$ and $e_1 - B(g_1, g_2) = 2a-4b > 0$;
- $e_2 - e_1 = 2b > 0$ and $e_2 \le 2a + 4b < 4a$;
- on the lower cell and the wall, $F_0 = a - 3b + 3c > 4a$, and $F_0 \le 4a + \frac{3}{2}b < \frac{19}{4}a$;
- on the upper cell, $F_0 = 3a + c > 4a$ and $F_0 \le 4a + 3b < \frac{11}{2}a$.

Part (b) follows from (a), and (c) from $2b < a$ and $c > a + b$. □

Mates and graphs. A *theta mate* is a positive definite quadratic lattice (L', Q) of rank four with $\theta_{L'} = \theta_{L_1}$. The theta series determines the determinant, so $\det G_{L'} = (27D)^2$. For a mate L', let R' be the span of its vectors with $Q \le 4a$. By the recognition theorem for C3 (Section 6), R' is isometric to R_0, R_1 or R_2. We call L' a *graph-1 mate* if $R' \cong R_1$. The other mates are treated in Section 6. By Corollary 7.5, $R_1 = \mathbb{Z}g_1 + \mathbb{Z}g_2$ is binary.

Theorem 7.2 (C3, graph 1)**.** *Let (a, b, c) be a primitive seed of the closed chamber* (4) *with $D \equiv 2 \pmod 3$ and $a > 10$. Then every graph-1 theta mate is isometric to L_1. The statement covers:*

- *the open lower cell;*
- *the open upper cell, together with the edge $c = a + 3b$;*
- *the tie wall $2c = 2a + 3b$, where fourth basis vectors with $Q(v) = F_0$ are kept.*

The proof is computer-assisted, and runs as follows.

- §7.2 proves the first excess and bounds the remaining basis vector. This step is computer-assisted: its positivity statements are exact Bernstein certificates.
- §7.4 classifies the rank-three span by an exact symbolic computation.
- §7.5 excludes six alternative spans by complete per-seed certificates.
- §7.6 completes the known span on both cells and the wall, with no unresolved branch.
- §7.7 gives an end-to-end check, independent of the completion engine, at every seed with $a \le 20$.
- §7.8 assembles the proof, and §7.9 states the independent checks.

Where the hypotheses enter. $D \equiv 2 \pmod 3$ and $a > 10$ are used only to set aside the exceptions of Proposition 7.11. Every other step holds at every primitive seed of the closed chamber.

- Seeds with $D \not\equiv 2 \pmod 3$ are covered by Section 3.
- The exceptions with $a \le 10$ are covered by the finite register (Remark 4.2).

7.2. **The first excess.**

Lemma 7.3. $Q(X, U) = (a - b)\,\mathrm{N}(X) + b\,\mathrm{N}(X + U) + (c - b)\,\mathrm{N}(U)$.

Proof. Expand $\mathrm{N}(X+U) = \mathrm{N}(X) + \mathrm{N}(U) + T(X,U)$. □

Notation for the proof.

- Write $A' = a - b$, $B' = b$ and $C' = c - b$. All three are positive on C3, by Lemma 7.1.
- For $v = (X, U)$ write $(n_1, n_{12}, n_2) = (\mathrm{N}(X), \mathrm{N}(X+U), \mathrm{N}(U))$.
- Then $Q(v_4) = A' + B' + 3C'$, with triple $(1,1,3)$, and $Q(h) = 3A' + 4B' + C'$, with triple $(3,4,1)$.

Theorem 7.4 (First excess)**.** *Let (a,b,c) be any real point of* C3*, and $v \in L_1 \setminus \Pi$. Then $Q(v) \geq F_0$, with equality exactly for:*

- *$v = \pm v_4$, when $2c \leq 2a + 3b$;*
- *$v = \pm h$, when $2c \geq 2a + 3b$.*

Proof. We use $F_0 < 6a$, $A' > B' > 0$, $4A' > 3B'$ and $C' > 0$ (Lemma 7.1). The cases below are exhaustive, because $n_2 \in \{0, 1, 3, 4, \dots\}$.

(i) A zero entry. $\det M = -1$ and $\det(M+I) = 1$, so both matrices are invertible modulo 3.

- If $U = 0$, then $X \equiv 0$ and $Q = 9a\,\mathrm{N}(X/3) \geq 9a$.
- If $X = 0$, then $U \equiv 0$ and $Q \geq 9c$.
- If $U = -X$, then $(M+I)X \equiv 0$, so $X \equiv 0$ and $Q = 9(a + c - 2b)\,\mathrm{N}(X/3)$.

In each case $v \neq 0$ gives $Q(v) > 6a > F_0$.

(ii) $n_1, n_{12} \geq 1$ and $n_2 \geq 3$. Then $Q \geq A' + B' + 3C' = Q(v_4) \geq F_0$.

- Equality forces the triple $(1,1,3)$ and $Q(v_4) = F_0$.
- The triple means that X and $X+U$ are units of N with $\mathrm{N}(U) = 3$. This gives twelve vectors, and a direct check shows that only $\pm v_4$ satisfy $U \equiv MX$.

(iii) $n_2 = 1$ and $n_1, n_{12} \geq 1$. Then $Q = A'n_1 + B'n_{12} + C'$, and there are five subcases.

- $n_1 = 1$. For $X = \pm(1,0)$, $MX \equiv \pm(1,1)$, which is congruent to no unit. For the other four units X, MX is itself a unit. Units are pairwise incongruent modulo 3, so $U = MX$ and $v \in \Pi$.
- $n_1 = 3$. Then $X \in \{\pm(1,1), \pm(-1,2), \pm(2,-1)\}$, and U is the unique unit congruent to MX.
  - For $X = \pm(-1,2)$ this gives $v \in \Pi$.
  - For the others it gives $\pm h$, and $\pm(2,-1,1,-1)$ with $Q = 3a + 3b + c > Q(h)$.
- $n_1 \geq 4$ *and* $n_{12} \geq 3$. Then $Q \geq 4A' + 3B' + C' > Q(h)$, since $A' > B'$.
- $n_1 = 4$ *and* $n_{12} = 1$. Then $X = 2\varepsilon$ with ε a unit, which forces $U = -\varepsilon$. Then $(M-I)\varepsilon \equiv 0$, which is impossible since $\det(M-I) = -1$.
- $n_1 \geq 7$ *and* $n_{12} = 1$. Then $Q \geq 7A' + B' + C' > Q(h)$, since $4A' > 3B'$.

So in case (iii), $Q \geq Q(h) \geq F_0$, with equality only for $\pm h$ when $Q(h) = F_0$. □

Corollary 7.5. *At every point of* C3*:*

- *every vector of L_1 with $Q < F_0$ lies in $L_1 \cap \Pi = \mathbb{Z}g_1 + \mathbb{Z}g_2$, so $R_1 = \mathbb{Z}g_1 + \mathbb{Z}g_2$;*
- *the series $\theta_{L_1} - \theta_{R_1}$ begins with $2q^{F_0}$ off the tie wall, and with $4q^{F_0}$ on it.*

Proof. Theorem 7.4, together with $e_1, e_2 < 4a < F_0$ (Lemma 7.1). □

We call n an *outside value* if the coefficient of q^n in $\theta_{L_1} - \theta_{R_1}$ is positive.

Remark 7.6 (Computational confirmation, not used in the proof). By Lemma 7.3, $Q \geq \frac{a}{2}\mathrm{N}(X) + a\,\mathrm{N}(U)$ on the closed chamber. So every vector with $Q < 16a$ has $|x|, |y| \leq 6$ and $|u|, |w| \leq 4$.

- *The catalogue.* This box holds 1,520 nonzero vectors of L_1. Of these, 1,446 lie outside Π, with 644 distinct affine forms.
- *The faces.* For every form, the difference between the form and F_0 is nonnegative at the vertices of both closed cells, and its zero faces in C3 are exactly those of Theorem 7.4.
- *The seeds.* A per-seed check at all 5,320 integer seeds of the closed chamber with $a \leq 40$ (no primitivity filter) agrees.

The catalogue is complete for $Q < 16a$ on the closed chamber. It also serves as the value catalogue of the later steps [`dep:C3G1-A1`].

7.3. **The span S and the fourth basis vector.** Let L' be a graph-1 mate, and fix an isometry $R_1 \to R'$. Let g_1' and g_2' be the images of g_1 and g_2, so that $Q(g_i') = e_i$ and $B(g_1', g_2') = B(g_1, g_2)$.

Lemma 7.7 (Primitivity)**.** (a) *$L' \cap \mathbb{R}R' = R'$. Every vector of $L' \setminus R'$ has $Q \geq F_0$. Exactly two of them have $Q = F_0$ off the tie wall, and exactly four on it.*

(b) *Let $h' \in L' \setminus R'$ with $Q(h') = F_0$; on the wall, h' may come from either pair. Then $S = \mathbb{Z}g_1' + \mathbb{Z}g_2' + \mathbb{Z}h'$ is primitive in L'.*

(c) *$|B(g_i', h')| \leq e_i$ for $i = 1, 2$. The vectors g_1', g_2', h' realise the successive minima e_1, e_2, F_0 of L' and of S.*

Proof. (a) Suppose $L' \cap \mathbb{R}R' \neq R'$.

- Lemma 5.5 gives $z \in (L' \cap \mathbb{R}R') \setminus R'$ with $Q(z) \leq (e_1 + e_2)/4 = (a + c)/2 < 4a$. This contradicts the definition of R'.
- So $L' \cap \mathbb{R}R' = R'$. Since $R' \cong R_1$, $\theta_{L'} - \theta_{R'} = \theta_{L_1} - \theta_{R_1}$, and Corollary 7.5 gives the rest.

(b) Otherwise, Lemmas 5.5 and 7.1 give $z \in (L' \cap \mathbb{R}S) \setminus S$ with $Q(z) \leq (e_1 + e_2 + F_0)/4 < F_0$. Since $z \notin S \supseteq R'$, part (a) gives $Q(z) \geq F_0$, a contradiction. Nothing about the second pair on the wall is used.

(c) We have $h' \pm g_i' \notin R'$, so $Q(h' \pm g_i') \geq F_0$, which is the inequality. For the minima:

- the only vectors of L' with $Q < e_2$ are $\pm g_1'$, as in R_1, a reduced binary lattice;
- every vector with $Q < F_0$ lies in R', which has rank two.

□

Lemma 7.8 (Minkowski's bound in rank three)**.** *Let S be a lattice of rank three whose successive minima, measured by Q, are $q_1 \le q_2 \le q_3$. Then $\det G_S \ge 4q_1q_2q_3$. In particular $\det G_S \ge 4e_1e_2F_0$ for the lattice S of Lemma 7.7.*

Proof. The inequality used. Let Λ have rank n and Gram matrix G, with squared successive minima $m_1 \le \cdots \le m_n$ for $\|x\|^2 = x^{\mathsf{T}}Gx = 2Q(x)$. Minkowski's second theorem, in Hermite's form, states $m_1 \cdots m_n \le \gamma_n^n \det G$.

Its proof.

- Let s_i realise the minima. Let τ scale the i-th Gram–Schmidt direction of (s_i) by $m_i^{-1/2}$.
- Let $x \ne 0$ have highest Gram–Schmidt index k, and let $j \le k$ be minimal with $m_j = m_k$. Then x is independent of $s_1, \dots, s_{j-1}$, so $\|x\|^2 \ge m_k$, and hence $\|\tau x\|^2 \ge \|x\|^2/m_k \ge 1$.
- Hermite's inequality for $\tau\Lambda$ gives $1 \le \gamma_n(\det G/\prod_i m_i)^{1/n}$.

The bound. Take $\gamma_3^3 = 2$ (Gauss) and $m = (2q_1, 2q_2, 2q_3)$. Then $8q_1q_2q_3 \le 2\det G_S$. For the lattice S of Lemma 7.7, part (c) gives $(q_1, q_2, q_3) = (e_1, e_2, F_0)$. □

Proposition 7.9. *$L' = S \oplus \mathbb{Z}v$ for any $v \in L'$ that generates L'/S. Choose v shortest in its coset. Then*

$$F_0 \le f := Q(v) \le \Phi := \frac{(27D)^2}{8e_1e_2F_0} + \frac{e_1 + e_2 + F_0}{4},$$

and $f > F_0$ off the tie wall.

- *For $s \in \{g_1', g_2', h'\}$, the pair norm $t_s := Q(v) + Q(s) - |B(v, s)|$, which is the norm of one of $v \pm s$, satisfies $f \le t_s \le f + Q(s)$.*
- *f and every t_s are outside values.*
- *On the closed chamber, $\Phi < 11a$ and $\Phi + F_0 < 16a$. Hence $f < 11a$ and $t_s < 16a$.*

Proof. The decomposition and the bound on f.

- S is primitive of rank three (Lemma 7.7), so $L'/S \cong \mathbb{Z}$.
- The component of v orthogonal to S has $\|\cdot\|^2 = (27D)^2/\det G_S$. Rounding the S-component by Lemma 5.5, and applying Lemma 7.8, gives $f \le \Phi$.
- Since $v \notin R'$, $f \ge F_0$. Off the wall, the only vectors of $L' \setminus R'$ with $Q = F_0$ are $\pm h' \in S$, so $f > F_0$.
- Minimality in the coset gives $|B(v, s)| \le Q(s)$.
- The vectors v and $v \pm s$ lie outside $S \supseteq R'$. Since $\theta_{L'} - \theta_{R'} = \theta_{L_1} - \theta_{R_1}$, their norms are outside values.

The two windows. Put $a = 1$.

- On each closed cell, F_0 is a single affine form, and $e_1, e_2, F_0 > 0$.
- So $\Phi < K$ is equivalent to

$$P_K := 8e_1e_2F_0\Big(K - \frac{e_1 + e_2 + F_0}{4}\Big) - 729(\gamma - \beta^2)^2 > 0,$$

  a quartic in (β, γ). And $\Phi + F_0 < 16$ is the same condition with $K = 16 - F_0$.
- Write P_{11} and P_{16-F_0} in the degree-four Bernstein basis of each closed triangle. All fifteen coefficients are positive, in exact arithmetic and without subdivision [`dep:C3G1-A1`]. Since the basis is nonnegative and sums to one, both polynomials are positive on the closed cells.

Finally, $\max(e_1, e_2, F_0) = F_0$, so $t_s \le f + F_0 < 16a$. □

Remark 7.10 (Numerical observation, not used). On a 600×600 grid per cell, in floating point:

- on the upper cell, $\max \Phi \approx 10.1425a$ (near the edge $c = a + 3b$), and $\max(\Phi + F_0) \approx 15.6147a$ (at the vertex $(\frac{1}{2}, \frac{5}{2})$);
- on the lower cell, the maxima are about $8.744a$ and $13.274a$.

These are not bounds.

7.4. **The rank-three span.** Replacing h' by $-h'$, we may assume $\beta_1 := B(g_1', h') \ge 0$. Put $\beta_2 := B(g_2', h')$. Then

$$G_S = \begin{pmatrix} 2e_1 & B(g_1, g_2) & \beta_1 \\ B(g_1, g_2) & 2e_2 & \beta_2 \\ \beta_1 & \beta_2 & 2F_0 \end{pmatrix}.$$

The known pairs. In L_1, the span $S_1 := \mathbb{Z}g_1 + \mathbb{Z}g_2 + \mathbb{Z}v_4 = \mathbb{Z}g_1 + \mathbb{Z}g_2 + \mathbb{Z}h$ (note $h - v_4 = g_2$) has this form, with

$$(\beta_1, \beta_2) = (a - 2b,\ a + 2b - 3c) \qquad \text{for } h' = v_4,$$
$$(\beta_1, \beta_2) = (b + c,\ 3a + 4b - c) \qquad \text{for } h' = h.$$

Also $\det G_{S_1} = 54De_1$. These are the *known pairs*:

- the first on the lower cell;
- the second on the upper cell;
- either on the wall, where they are the two labelings of S_1.

Proposition 7.11 (computer-assisted)**.** *Let L' be a graph-1 mate at a seed of the closed chamber. Then (β_1, β_2) is a known pair, unless the seed is one of* 42 *seeds at which, in total,* 66 *alternative pairs pass every test. These are isolated points: no alternative survives along a segment or on a two-dimensional region. Exactly six of the* 42 *seeds are primitive with $D \equiv 2 \pmod 3$ and $a > 10$, each with one alternative:*

Cell	*Seed* (a,b,c)	*Alternative* (β_1,β_2)
lower	$(12,5,19)$	$(2,-15)$
lower	$(36,16,59)$	$(4,-45)$
upper	$(12,5,27)$	$(11,29)$
upper	$(15,7,33)$	$(16,-26)$
upper	$(18,5,31)$	$(33,26)$
upper	$(24,11,53)$	$(25,-42)$

The other 60 *points lie at seeds with* $D \not\equiv 2 \pmod 3$ *or* $a \le 10$. *The complete list of the* 66 *points is pinned in [*`dep:C3G1-A2`*].*

Method. The branches. The pair norms $F_{13} := Q(h'-g_1') = e_1+F_0-\beta_1$ and $F_{23} := e_2+F_0-|\beta_2|$ are outside values in $[F_0, e_i+F_0]$, and $e_i+F_0 < \frac{19}{2}a$. So each equals, at the seed, one of the 45 affine forms of catalogue vectors outside Π whose minimum on the closed chamber is at most $12a$. A *branch* is a choice $(F_{13}, F_{23}, \operatorname{sign}\beta_2)$ on a stratum (lower cell, upper cell or wall). That gives 4,050 branches per stratum.

The domains. On a branch, G_S is affine in (a,b,c). Its domain is the closed stratum cut by $F_0 \le F_{i3} \le e_i + F_0$.

The necessary conditions, imposed exactly on the 97 test vectors $x = kh' + ig_1' + jg_2'$ with $k \in \{1,2\}$, $|i|, |j| \le 3$ and $x \ne h'$:

- *minimality:* $Q(x) \ge F_0$, since $x \notin R'$;
- *value:* each $Q(x) < 12a$ is an outside value, enforced by exact cuts along the lines $Q(x) = F$ for the 45 forms;
- *count:* at each value below $12a$, the test vectors and h', each counted with its negative, number at most the coefficient of $\theta_{L_1} - \theta_{R_1}$ there.

Strictness. Strict conditions are tracked exactly: the open edges, and $Q(x) > F_0$ and $t > F_0$ off the wall. A piece on which a strict condition vanishes identically is discarded.

The survivors. Of the 272, 290 and 272 branches with nonempty domain, 17, 79 and 28 survive. They give:

- the known pair on each whole closed stratum, with both labelings on the wall;
- pieces of other branches on which the Gram coincides with the known one;
- the 66 points.

Each point passes a complete exact test at its seed ($\theta_S - \theta_{R_1} \le \theta_{L_1} - \theta_{R_1}$ coefficientwise through $12a$), so all of them are genuine per-seed candidates.

The wall. The branches enumerate every labeled Gram, whichever shortest vector is called h'. So the result does not depend on that choice.

A check. At all 562 primitive seeds of the closed chamber with $a \le 20$, an exact per-seed enumeration of the possible (β_1,β_2), with no box, agrees with the surviving pieces: none is missing and none is extra. □

7.5. **The six alternative spans.**

Proposition 7.12 (complete finite certificates)**.** *At each seed of the table in Proposition 7.11, the alternative span S has no completion. That is, there is no lattice $S \oplus \mathbb{Z}v$ with Gram determinant $(27D)^2$ such that:*

- *$f = Q(v) \in [F_0, 11a)$ is an outside value;*
- *every pair norm $t_s \in [f, f + Q(s)]$ is an outside value.*

By Proposition 7.9, every completion of a graph-1 mate would be such a lattice.

Proof. At each seed we enumerate:

- every outside value $f \in [F_0, 11a)$;
- every triple of outside values $t_s \in [f, f + Q(s)]$;
- every sign vector, with $u_s = \pm(Q(s) + f - t_s)$.

We compute $\det \left(\begin{smallmatrix} G_S & u \\ u^{\mathsf{T}} & 2f \end{smallmatrix} \right)$ exactly. All values used are below $16a$, where the catalogue of Remark 7.6 is complete. No case has determinant $(27D)^2$:

Seed	$\det G_S$	values of f	cases	solutions
$(12, 5, 19)$	338,772	21	310,408	0
$(36, 16, 59)$	9,562,488	23	459,196	0
$(12, 5, 27)$	591,888	13	146,021	0
$(15, 7, 33)$	1,053,804	13	140,396	0
$(18, 5, 31)$	1,353,348	17	185,128	0
$(24, 11, 53)$	4,373,952	14	182,430	0

The control. The same enumeration was applied to the known span S_1 at the same six seeds.

- It finds $6, 4, 4, 6, 2, 6$ determinant solutions, counting v and $-v$.
- Of these, $1, 1, 2, 3, 1, 3$ are theta-equal through $16a$, up to sign.
- Each theta-equal one is isometric to L_1, by an exact isometry, and to neither L_0 nor L_2.
- The others fail theta.

The data are pinned in [`dep:C3G1-A3a`]. □

7.6. **Completion of the known span.** Assume the hypotheses of Theorem 7.2. By Propositions 7.11 and 7.12, the span S of a graph-1 mate has a known Gram.

- So $(g_1', g_2', h') \mapsto (g_1, g_2, s_3)$, with $s_3 \in \{v_4, h\}$, is an isometry onto S_1.
- $L' = S \oplus \mathbb{Z}v$ is then determined by $f = Q(v)$ and $u_i = B(s_i', v)$.

It remains to find every such completion, and to identify each one. This is done by the completion engine of Section 11, in three configurations:

Config.	Closed domain (open conditions)	Frame	Window for f
lower	lower triangle ($\beta \neq 0, \frac{1}{2}$; $c > a + b$; off the wall)	(g_1, g_2, v_4)	$a - 3b + 3c \leq f \leq 11a$
upper	upper triangle ($\beta \neq 0, \frac{1}{2}$; off the wall)	(g_1, g_2, h)	$3a + c \leq f \leq 11a$
wall	segment $2c = 2a + 3b$ ($0 < \beta < \frac{1}{2}$)	both frames	$F_0 \leq f \leq 11a$

The edge $c = a + 3b$ is included in the upper configuration.

The universe.

- *The target forms* are the affine forms of the nonzero vectors of L_0 whose minimum on the configured closed domain is below $16a$. They were checked against the box of Remark 7.6, which contains every vector with $Q < 16a$ for either lattice. For each configuration, the target forms are exactly the forms of L_0 there: 80, 79 and 79 forms. Those of L_1 are the same, with equal multiplicities.
- *A unit* is a triple (f, t_1, t_2) of target forms. *A branch* adds t_3 and signs (σ_2, σ_3); $\sigma_1 = +1$ normalises $v \mapsto -v$.
- *The Gram of a branch.* With $u_i = \sigma_i(Q(s_i) + f - t_i)$, the Gram matrix $G = \left(\begin{smallmatrix} G_{S_1} & u \\ u^{\mathsf{T}} & 2f \end{smallmatrix} \right)$ is affine in (a, b, c). For $a = 1$, $P := \det G - 729(\gamma - \beta^2)^2$ is a polynomial of degree at most four in (β, γ).
- *The domain of a branch* is the configured closed domain, cut by the window for f and by $f \leq t_i \leq f + Q(s_i)$.
- *Coverage.* By Proposition 7.9, f and the t_i of a graph-1 mate are outside values below $16a$. Each is therefore the value at the seed of a target form, so every completion lies in the domain of a branch at its seed.
- The universe consists of every unit with nonempty domain and every branch within it.

How branches are decided (exact arithmetic throughout).

- *N:* P has no zero on the closed branch domain. This is certified by exact interval enclosures on dyadic boxes.
- *Line cuts.* For the 27 test vectors $x = x_1 s_1 + x_2 s_2 + x_3 s_3 + v$ with $x_i \in \{-1, 0, 1\}$:
  - $Q(x)$ is affine on the branch, and wherever $Q(x) < 16a$ it must equal a target form;
  - so the zeros of P lie on lines $Q(x) = F$ or in the region where $Q(x) \geq 16a$;
  - on each line, the rational zeros of P are found exactly, by the rational root theorem for an integer quartic, and the rest of the domain is certified zero-free.
- *The labels of the points found:*

  - *O:* outside the open domain;
  - *L:* fails the value test;
  - *P:* a candidate;
  - *R:* the branch has no point at all.
- *Families.* Suppose P vanishes identically on the domain (an identity family) or on a cut line (a one-dimensional family). Then a resolver imposes necessary count and short-shell conditions. Each surviving piece is:
  - *K:* identified with L_1 by an explicit integral basis map, checked as an identity of affine forms on the closed piece;
  - *P:* a single point;
  - *U:* unresolved, which is saved.

  A family point that the resolver excludes is dropped. It carries a count witness: a value below $16a$ taken by more test vectors than θ_{L_1} allows.
- *Candidates.* Each candidate is evaluated at its seed: the integer Gram, $\det = (27D)^2$, and theta through $16a$. A theta match is tested for isometry with L_0, L_1 and L_2 by an exact isometry search.

The result [`dep:C3G1-A3b`]. A branch may carry several labels, so a row of labels can sum to more than its branch count.

Config.	units	branches	N	R	O	L	K	P	U
lower	31,214	4,170,988	3,980,035	185,507	5,322	146	23	11	0
upper	30,077	4,120,168	4,022,248	91,759	5,304	249	97	680	0
wall	54,662	7,293,008	7,097,361	185,711	9,379	269	21	306	0
total	115,953	15,584,164							0

No branch is unresolved. The distinct candidates (by configuration, seed and Gram) come out as follows:

Config.	candidates	theta-equal through $16a$	isometric to L_1
lower	19	19	19
upper	208	198	198
wall	83	83	83

- The ten upper candidates that are not theta-equal lie at the seeds $(3, 1, 5)$ (eight), $(3, 1, 6)$ and $(7, 2, 12)$.
- Every theta-equal candidate is isometric to L_1, by an exact isometry, and every K piece is L_1, by an explicit integral basis map.
- No candidate is theta-equal without being isometric to a known lattice.

Controls.

- *K1:* the K pieces of the known lattice contain every primitive seed with $a \leq 40$ of the open domains: 928, 3,394, and 122 for each wall frame.

- *K2:* at 29 control pairs (frame, seed), the branch hits equal an independent brute-force set of determinant solutions.
- *K2′:* at the same pairs, every theta-matching completion found by brute force (6, 15 and 35, up to sign) is reached by a K piece or a P record.
- The branch domains agree exactly with the configured domains.
- A second implementation of the points stage and of K2′ gives the same candidates and verdicts.

Remark 7.13. The engine's generic sufficient condition $f_{\max} + \max_i Q(s_i) \le 16a$ does not hold literally here: $11a + F_0$ can approach $16.5a$. It is not needed. For an actual mate, $t_i \le f + Q(s_i) \le \Phi + F_0 < 16a$ by Proposition 7.9. So every t_i of an actual mate is a target value, and no actual mate needs a branch beyond $16a$.

7.7. **An end-to-end check.** The family exclusions above drop points with a count witness. K2′ guards against losing a mate that way, but only at the control seeds. The following check extends it to every small seed. It is an independent consistency check, not a dependency of Theorem 7.2.

Proposition 7.14 (computed)**.** *The check covers every primitive seed with* $a \le 20$ *of the three configurations:* 94 *lower,* 438 *upper and* 30 *wall seeds, which are exactly the* 562 *primitive seeds of the closed chamber with* $a \le 20$*. It also covers the two control seeds* $(36, 16, 59)$ *and* $(24, 11, 53)$*. That makes* 594 *pairs (frame, seed). At these pairs:*

- *an exact brute-force enumeration of all completions of the known span finds* 942 *completions that are theta-equal through* $16a$*, up to sign;*
- *the engine reaches each of them, always within a K piece;*
- *each is isometric to* L_1*, by an exact isometry, and to neither* L_0 *nor* L_2*.*

Config.	pairs	theta matches	reached	missing	isometric to L_1 only
lower	95	95	95	0	95
upper	439	708	708	0	708
wall	60	139	139	0	139
total	594	942	942	0	942

The data are pinned in [`dep:C3G1-A3c`]. Outside the brute-force set, the only engine completions are the ten upper candidates that fail theta.

A consequence. Steps §7.2–§7.5 and this check use code independent of the completion engine, and the brute force is complete at each seed. So for the seeds of Theorem 7.2 with $a \le 20$, the conclusion does not depend on the engine.

7.8. **Proof of Theorem 7.2.** Let L' be a graph-1 mate at a primitive seed of the closed chamber with $D \equiv 2 \pmod 3$ and $a > 10$.

- By Lemma 7.7 and Proposition 7.9, $L' = S \oplus \mathbb{Z}v$, with S primitive and f and the t_s in the stated windows.
- By Proposition 7.11, the hypotheses on D and a leave two possibilities for S:
  - an alternative span from the table, which has no completion (Proposition 7.12);
  - a known Gram, which is the remaining case.
- With a known Gram, the basis $(g_1', g_2', h', \pm v)$ gives a branch of the configuration for the seed's stratum, with the seed in its domain. On the wall, either frame may be used.
- That branch is not decided N, R, O or L at the seed, and family points are excluded only by necessary conditions. Since $U = 0$, the seed lies in a K piece or is a P candidate.
  - In a K piece, the explicit basis map identifies L' with L_1, since two lattices with equal Gram matrices are isometric.
  - A P candidate is theta-equal, being a mate, and every theta-equal candidate is isometric to L_1.

So $L' \cong L_1$. □

7.9. **Independent checks.** The following checks used independently written code and executed no producer code. They are stated with their exact scope.

The first excess and the windows. Theorem 7.4, Lemmas 7.7 and 7.8 and Proposition 7.9 were reviewed against the reviewer's own certificates:

- a complete catalogue below $12a$, with exact cell queries, for the first excess and its multiplicities;
- an independent derivation of the factor 4 in Lemma 7.8;
- sharper exact maxima, which certify both windows.

The 5,320-seed run and the complete $16a$ catalogue of Remark 7.6 were not replayed.

The rank-three span.

- *Rebuilt and checked completely:*
  - the branch index: the $272 + 290 + 272$ branches with nonempty domain;
  - all 808 saved pieces;
  - all 138 alternative point records, by complete enumeration below $12a$;
  - the 124 branches that contain a non-boundary piece, shown by exact solver queries to have no survivor outside the saved pieces.
- *Sampled:*
  - of the remaining routine exclusions, 64 were checked the same way, and the other 646 were not individually replayed;
  - the per-seed check of Proposition 7.11 was recomputed at 64 of its 562 seeds, and the other per-seed sets were used as data.

The alternative spans.

- The six completion universes and their controls were rebuilt and checked in full, with a different enumeration: an exact quadratic in the third pairing.
- No determinant solution was found.
- Every theta match of the controls was given an explicit ambient basis of L_1.

The completion of the known span.

- *Rebuilt and checked completely:*
  - all 115,953 units and 15,584,164 branches, with each closed-domain choice present exactly once, the per-unit label counts in agreement, and no U;
  - every one of the 21,543 nonroutine branches;
  - all 134,216 listed point records and all 175 known pieces;
  - all 310 candidates, by exact theta counts through $16a$, with explicit integral basis maps for the 300 matches;
  - the 29 control cases.
- *Sampled:* the routine N and R decisions, in a fixed sample of 64 N and 64 R decisions per configuration.
  - All 192 R queries and 181 of the 192 N queries passed exact solver checks.
  - The 11 N queries that timed out were closed by separately predeclared exact Bernstein certificates.
- *So this audit is not a decision-by-decision replay of the routine decisions.* For those, the theorem relies on the completion engine (Section 11), and for $a \le 20$ also on Proposition 7.14.

Replays. Every producer job behind the results of this section was replayed from a fresh mirror of its pinned inputs, and each reproduces:

- the first excess and the windows;
- the rank-three classification and its post hoc re-check;
- the six alternative spans;
- the six engine runs of §7.6 and its fourteen checks;
- the end-to-end check.

Nine sizing and pilot runs of §7.6, which enter no result, were not replayed.

Review. The internal review accepted the chain §§7.2–7.7, including the tie wall with $Q(v) = F_0$ kept.

8. Chamber C2

Throughout this section (a, b, c) is a seed of the closed chamber

$$\text{C2}: \qquad 0 < 2b < a, \qquad a + 3b < c \le 3a - b, \tag{5}$$

that is $0 < \beta < \frac{1}{2}$ and $1 + 3\beta < \gamma \le 3 - \beta$. The wall $c = 3a - b$ belongs to C2, and the edge $c = a + 3b$ belongs to C3. Coordinates and notation are those of Section 5; $\Delta = (27D)^2$.

The short vectors. On C2 we have $e_1 < e_2 \le 4a < e_3$: $e_2 \le 4a$ is $c \le 3a - b$, and $e_3 > 4a$ is $c > a + 3b$. The vectors of L_j with $Q \le 4a$ are exactly $\pm g_1, \pm g_2$, so $R_j = \mathbb{Z}g_1 + \mathbb{Z}g_2$ is binary, with

$$B_j := |B(g_1, g_2)| = c - a, \quad c + 3b - a, \quad c - a - 3b \qquad (j = 0, 1, 2).$$

A theta mate K has exactly the four vectors $\pm g_1', \pm g_2'$ with $Q \le 4a$, $Q(g_i') = e_i$. Its complete $4a$ span is $R = \mathbb{Z}g_1' + \mathbb{Z}g_2'$, of rank two since $\pm g_2'$ are the only vectors of norm e_2. Change the sign of g_2' so that $B := B(g_1', g_2') \ge 0$.

8.1. **Recognition of the binary span.**

Theorem 8.1 (Recognition in C2)**.** *Let (a, b, c) be a primitive seed of the closed chamber* (5) *with $D \equiv 2 \pmod 3$, and K a theta mate of L_0. Then $B \in \{B_0, B_1, B_2\}$, that is, the complete $4a$ span of K is isometric to R_0, R_1 or R_2. [`dep:C2`]*

Proof. The reduction. The vector $g_2' - g_1'$ is not in $\mathbb{Z}g_1'$, so $Q(g_2' - g_1') = e_1 + e_2 - B \ge e_2$, that is $0 \le B \le e_1$. Its norm is at most $e_1 + e_2 = 2a + 2c \le (8 - 2\beta)a < 8a$, so it is a value of θ_L, and $B = e_1 + e_2 - F$ for a catalogue form F: one branch per form.

The symbolic classification. The conditions (C1)–(C3) of Section 5.2 (values, the short-shell rule for e_1 and e_2, counts) were applied to every vector of a box, on exact closed domains in the (β, γ)-plane.

- *At window* $12a$ (box of radius 3), the survivors on the generic chamber are the three known values and one alternative, $B = c + 2b - 2a$. The alternative survives only on three segments: $c = 2a$ for $0 < \beta < \frac{1}{3}$; $2c = 7(a - b)$ for $\frac{1}{5} < \beta < \frac{5}{13}$; and $2c = 7a - 6b$ for $\frac{1}{4} \le \beta < \frac{5}{12}$. On the segment $\beta = \frac{1}{5}$ it coincides with B_2.
- *At windows* $16a$, $20a$ *and* $24a$ (the same code, with larger box and window), the three segments shrink to isolated points. At $24a$ the alternative remains only at the four seeds $(3, 1, 7)$ and $(8, 3, 19)$, with $D \equiv 2 \pmod 3$, and $(4, 1, 8)$ and $(7, 2, 14)$, with $D \equiv 1 \pmod 3$. Where it survives it is a different span: its B is not a known B_j there.

The two in-scope seeds have $a \le 10$. There the register [`dep:REG`] proves exact-three, so the span of every mate is a known span (Remark 4.2). □

Cross-validation and disclosure. An exact enumeration of all admissible B at every primitive seed of the closed chamber with $a \le 10$ (124 seeds), with no box, agrees with the symbolic record at the windows $12a$, $16a$ and $24a$: sound and sharp at 124/124. The $12a$ classification and its cross-validation were predeclared. The runs at $16a$, $20a$ and $24a$, which reduce the alternative to four seeds, were additional; they are exact necessary tests, but they are not predeclared evidence.

8.2. **The forced third vector and the rank-three span.** Put $F_0 = e_3 = 3a + c - 3b$ for graphs 0 and 2, and $F_0 = 3a + c$ for graph 1.

Lemma 8.2 (The forced third vector)**.** *Let K be a theta mate at a seed of (5), the wall included, whose complete $4a$ span R is isometric to R_j. Then:*

(i) *every vector of K with $Q < F_0$ lies in R;*
(ii) *exactly two vectors of K outside R have $Q = F_0$, a pair $\pm h$;*
(iii) *$M_3 := K \cap (\mathbb{R}R + \mathbb{R}h) = \mathbb{Z}g_1' + \mathbb{Z}g_2' + \mathbb{Z}h$, with successive minima e_1, e_2, F_0 attained by g_1', g_2', h, and h is a shortest vector of K outside R;*
(iv) *$|B(g_i', h)| < e_i$ for $i = 1, 2$.*

Proof. R is primitive (Lemma 5.6). Since $\theta_K = \theta_L$ and $\theta_R = \theta_{R_j}$, the number of vectors of K outside R at each value n is $\theta_L(n) - \theta_{R_j}(n)$, the number of vectors of L_j outside R_j at n.

- *The catalogue.* For every vector of the complete catalogue of L_j below $12a$ outside R_j, its form τ satisfies $\tau - F_0 \geq 0$ at the three vertices of the closed chamber, hence on all of it. Equality $\tau \equiv F_0$ holds for one pair $\pm h_j$ only. For every other outside vector, $\tau - F_0$ vanishes only at vertices, or on the edge $\beta = 0$. These sets contain no seed, since every seed has $0 < 2b < a$.
- (i) and (ii) follow. For (iii), the vectors of M_3 below F_0 lie in R, so M_3 has successive minima e_1, e_2, F_0, attained by g_1', g_2', h. In rank three, successive-minimum vectors form a basis, and M_3 is primitive by definition.
- (iv) $h \pm g_i'$ lie outside R and differ from $\pm h$, so $F_0 + e_i \pm B(g_i', h) > F_0$.

□

In L_j itself the known rank-three span $M_{3,j} = \mathbb{Z}g_1 + \mathbb{Z}g_2 + \mathbb{Z}h_j$ has, with $B_{12} = B_j$,

$$(B_{13}, B_{23}) = (3a - b - c,\ c - 4b), \quad (b + c,\ 3a + 4b - c),$$
$$(c - 2b,\ 3a - 2b - c) \qquad (j = 0, 1, 2).$$

Proposition 8.3 (Stage 1: the rank-three span)**.** *For each graph j, the possible Gram matrices of M_3 have $B_{12} = B_j$, $B_{13} = e_1 + F_0 - F_{13} \in [0, e_1]$ after the sign of h, and $B_{23} = \pm(e_2 + F_0 - F_{23})$ with $|B_{23}| \leq e_2$, for catalogue forms F_{13}, F_{23}. A symbolic classification at window $12a$, with generator norms (e_1, e_2, F_0) and the single pair at F_0 enforced through the counts, gives:*

- *$M_{3,j}$ survives on the whole closed chamber for each graph;*
- *no alternative survives on any two-dimensional region;*
- *genuine alternatives survive only on 73, 113 and 79 isolated points (graphs 0, 1, 2), and on 15, 23 and 15 segments.*

An exact per-seed enumeration at the 124 *seeds with* $a \le 10$*, for all three graphs, lies inside the symbolic prediction at all* 372 *seed–graph pairs.* *[dep:C2]*

Lemma 8.4 (The fourth vector)**.** *In the situation of Lemma 8.2,* $K = M_3 \oplus \mathbb{Z}v$ *for a vector* v*, shortest in its coset, with*

$$F_0 < f := Q(v) \le \frac{\Delta}{2 \det M_3} + \frac{e_1 + e_2 + F_0}{4}.$$

For $M_3 = M_{3,j}$ *this gives* $f \le 8.73a$ *on the closed chamber, and the smaller pair norm* $\min_{\pm} Q(v \pm s)$ *of each* $s \in \{g_1', g_2', h\}$ *is below* $14.73a$*. For an arbitrary* M_3*, Minkowski's bound* $\det M_3 \ge 4e_1e_2F_0$ *(Lemma 7.8; the minima are those of Lemma 8.2(iii)) gives* $f < 12.871a$ *and smaller pair norms below* $18.273a$*.*

Proof. M_3 is primitive of rank three, so $K/M_3 \cong \mathbb{Z}$. Rounding (Lemma 5.5) gives the bound. A vector outside M_3 has $Q \ge F_0$ by Lemma 8.2(i), and $Q = F_0$ is impossible because $\pm h \in M_3$; so $f > F_0$. The numerical bounds follow from exact interval arithmetic on a cover of the closed chamber by rectangles, using the cofactor expansion of $\det M_{3,j}$, or Minkowski's bound where needed [dep:C2]. □

8.3. **Stage 2: the completion.** *Case I: completing* $M_{3,j}$. A branch chooses the graph, a catalogue form f with $F_0 < f \le 9a$, catalogue forms $t_1 \in [f, f + e_1]$, $t_2 \in [f, f + e_2]$ and $t_3 \in [f, f + F_0]$ below $16a$ for the pair norms of v with g_1', g_2' and h, and the signs. There are 46,023 units and 5,675,452 branches with nonempty domains. Each branch is decided as in Section 5.4, on the closed chamber (5).

	N	R	O	L	P	K	U
Case I	5,512,393	140,969	21,087	681	508	188	0
Case II	—	23,580,897	8,107	152	37	0	0

TABLE 4. The C2 completion, with the boundary repair. Case I has 5,675,452 branches, 1,021 point records and 202 identified pieces. Case II has 23,589,188 branches on 53 segments. A branch can carry several letters.

The 390 distinct candidates of Case I all have $\det G = \Delta$. Of these, 377 agree with θ_L through $16a$, and each is isometric, by an exact isometry test, to some L_j (98, 151 and 128 copies of L_0, L_1, L_2, at seeds up to $a = 89$). The other 13 differ from θ_L first between $4.8a$ and $6.33a$, at seeds with $a \le 11$.

Remark 8.5 (The boundary repair). The first version of the engine skipped a restriction that held identically on the current segment or point. Such a restriction is a satisfied equality, and must be kept. In the repaired engine it is kept. Against the first run, the repaired run changes 6,340 branch

decisions, all from R to O, and adds four point records, at $(7, 3, 18)$ and $(6, 1, 16)$, whose Gram matrices already occur among the candidates. The 390 candidate rows are identical, and $U = 0$ in both runs. The repaired source is the one pinned in [`dep:C2`], and the later engines (C4, C3, C1, C0) inherit the repair.

Case II: the alternative rank-three spans. The 53 genuine alternative segments of Proposition 8.3 carry 23,589,188 branches, all decided with $U = 0$ (Table 4). Of their 37 candidate points, 34 agree with θ_L through $16a$; they are 5 distinct Gram matrices, at $(5, 1, 14)$, $(6, 1, 17)$ and $(15, 1, 41)$. All 5 are isometric to L_0. The other 3 candidates fail θ_L. At the 59 isolated alternative points with $D \equiv 2 \pmod 3$ and $a > 10$, an exact brute-force completion finds no candidate at all. The isolated points with $a \leq 10$ are closed by the register.

Controls. L_j is found as an identified K piece at all 124 seeds with $a \leq 10$ and at the six control seeds, for every graph. Brute force equals the symbolic hits at all 18 pairs (control seed, graph), and no candidate contradicts the register.

Theorem 8.6 (C2)**.** *Let (a, b, c) be a primitive seed of the closed chamber* (5) *with $D \equiv 2 \pmod 3$. Then every theta mate of L_0 is isometric to L_0, L_1 or L_2. [`dep:C2`]*

Proof. By Theorem 8.1, the span is $R \cong R_j$. By Lemma 8.2, $M_3 = \mathbb{Z}g_1' + \mathbb{Z}g_2' + \mathbb{Z}h$ is primitive, and by Proposition 8.3 it is $M_{3,j}$ unless the seed lies on one of the listed points or segments. By Lemma 8.4, $K = M_3 \oplus \mathbb{Z}v$ with v in the branch universe of Case I or Case II [`dep:UNIVERSES`]. Every branch is decided with $U = 0$; every K piece is an explicit L_j; and every candidate either fails θ_L or is isometric to some L_j. The isolated alternative points have no candidate when $a > 10$, and are covered by the register when $a \leq 10$. □

Remark 8.7 (What was regenerated independently). A reviewer regenerated the complete repaired Case I computation (46,023 units and 5,675,452 branches), with fresh exact theta and isometry tests of the candidates, 18 brute-force controls and all 372 seed–graph known-completion controls. The reviewer checked the Case II algorithm and its records (23,589,188 branches, 37 candidate records, 59 point jobs), but did *not* independently regenerate all Case II branch decisions, nor all two-dimensional Stage 1 certificates. For those parts the dependencies are the producer's records and a full fresh replay of the producer; the independent audit of record inherits them too. Theorem 8.6 is therefore a computer-assisted theorem with disclosed dependencies, not a result of two fully independent engines [`dep:C2`].

9. Chambers C1 and C0 below the tail

This section treats the two chambers

$$\text{C1}: \qquad 0 < 2b < a, \qquad 3a - b < c \le 3a + b, \tag{6}$$

$$\text{C0 below the tail}: \qquad 0 < 2b < a, \qquad 3a + b < c, \qquad 20c \le 81a + 30b. \tag{7}$$

In the (β, γ)-plane their closures are the triangle with vertices $(0, 3)$, $(\frac{1}{2}, \frac{5}{2})$, $(\frac{1}{2}, \frac{7}{2})$, and the quadrilateral with vertices $(0, 3)$, $(\frac{1}{2}, \frac{7}{2})$, $(\frac{1}{2}, \frac{24}{5})$, $(0, \frac{81}{20})$. The wall $c = 3a + b$ belongs to C1, and the tail wall $20c = 81a + 30b$ belongs to C0; the edge $c = 3a - b$ belongs to C2. Notation is that of Sections 5 and 8; $\Delta = (27D)^2$, and $B_j \in \{c - a,\ c + 3b - a,\ c - a - 3b\}$ are the binary pairings of Section 8.

Theorem 9.1 (C1 and C0 below the tail)**.** *Let (a, b, c) be a primitive seed with $D \equiv 2 \pmod 3$ and $a > 10$ in (6) or (7). Then every theta mate of L_0 is isometric to L_0, L_1 or L_2. The seeds with $a \le 10$ are covered by the register [`dep:REG`].*

The proof follows the pattern of C2, with one more step at the start. Above C2 the value e_2 exceeds $4a$, so the complete $4a$ span has rank one, and the structure is recovered from the first excess instead.

9.1. The first excess.

Proposition 9.2 (First excess)**.** *Let $L = L_j$, $j \in \{0, 1, 2\}$, at a point of (6) or (7). Then:*

1. *the minimum of L is e_1, attained only by $\pm g_1$;*
2. *the first coefficient at which θ_L exceeds $\theta_{\mathbb{Z}g_1}$ is at $e_2 = a + b + c$, where exactly one pair of vectors outside $\mathbb{Z}g_1$ occurs.*

This holds for every residue of D. [`dep:C1C0-S1`]

Proof (computer-assisted). The catalogue of L_j below $12a$ is complete: from $Q/a \ge \mathrm{N}(x, y) + \mathrm{N}(u, w) + \min(0, T/2)$ and $|T| \le \mathrm{N}(x, y) + \mathrm{N}(u, w)$, every vector of norm below $12a$ has $\mathrm{N}(x, y) + \mathrm{N}(u, w) < 24$. The exact lower envelope of the affine forms of all catalogue vectors outside $\mathbb{Z}g_1$, over the closed triangle and the closed quadrilateral, is e_2, attained by one pair. Its only tie is with e_3, on the excluded edge $\beta = \frac{1}{2}$, and $e_2 = e_1$ only on the excluded edge $\beta = 0$. On the line $\mathbb{Z}g_1$, the next vector $2g_1$ has $Q = 4e_1 > 12a$. □

A separate per-seed enumeration, in a larger box, at every primitive seed with $a \le 30$ (1,872 seeds of C1 and 4,261 of C0, three graphs each) agrees everywhere.

Corollary 9.3. *Every theta mate K has a unique pair $\pm g$ with $Q(g) = e_1$, and a unique pair $\pm h$ outside $\mathbb{Z}g$ with $Q(h) = e_2$, and no other vector outside $\mathbb{Z}g$ below e_2. The binary lattice $S_2 = \mathbb{Z}g + \mathbb{Z}h$ is primitive in K, and $|B(g, h)| \le e_1$. In L_j itself, $S_{2,j} = \mathbb{Z}g_1 + \mathbb{Z}g_2$ and $|B(g_1, g_2)| = B_j$.*

Proof. The first two statements follow from Proposition 9.2, since $\theta_K = \theta_L$. If S_2 were not primitive, Lemma 5.5 would give $z \in (K \cap \mathbb{R}S_2) \setminus S_2$ with $Q(z) \le (e_1 + e_2)/4 < e_2$, and $z \notin \mathbb{Z}g$, a contradiction. Finally $h \pm g$ lie outside $\mathbb{Z}g$, so $Q(h \pm g) \ge e_2$, that is $|B(g,h)| \le e_1$. □

9.2. **Recognition of the binary span.**

Theorem 9.4 (Binary recognition)**.** *Let (a,b,c) be a primitive seed with $D \equiv 2 \pmod 3$ and $a > 10$ in (6) or (7), and K a theta mate. Then $S_2 \cong S_{2,j}$ for some j; that is, after the sign of h, $B(g,h) \in \{B_0, B_1, B_2\}$. [dep:C1C0-D]*

Proof. Normalize $B = B(g,h) \ge 0$. The vector $h - g$ lies outside $\mathbb{Z}g$ and is not $\pm h$, so its norm $e_1 + e_2 - B$ is a value of θ_L larger than e_2; hence $0 \le B < e_1$. Since $e_1 + e_2 = 2a + 2c < 12a$, this gives one branch per catalogue form. The conditions (C1)–(C3) of Section 5.2 are applied, with the short-shell rule on e_1 and e_2 (single pairs, by Corollary 9.3), a box of radius 3, and windows $18a$, $24a$ and $36a$, on the exact closed domains.

- At window $36a$ no alternative survives on any two-dimensional region or any segment. Two segments, $\beta = \frac{1}{5}$ and $c = 6a - 3b$, keep a second value, but on each of them it is identically B_2.
- The alternatives survive only at 20 seeds. Nineteen have $a \le 10$ or $D \not\equiv 2 \pmod 3$.
- The remaining seed is $(11, 3, 43)$, with $D = 464$ and alternative $B = 27 = c + 2b - 2a$. An exact count test at that seed, through $48a$, excludes it. This test was additional, not predeclared.

□

Controls. The three known branches survive on the whole closure of both chambers at every window. An exact per-seed enumeration at all 3,138 primitive seeds with $a \le 24$ lies inside the symbolic prediction at every window (sound everywhere). As in C2, the alternatives occur on lines at small windows and thin out to isolated small seeds as the window grows.

9.3. **The forced third vector and the windows.** Put $F_0 = e_3 = 3a - 3b + c$ for graphs 0 and 2, and $F_0 = 3a + c$ for graph 1, as in C2.

Lemma 9.5 (The forced third vector)**.** *Let K be a theta mate at a seed of (6) or (7) with $S_2 \cong S_{2,j}$. Then:*

(i) *every vector of K outside S_2 has $Q \ge F_0$;*
(ii) *exactly one pair $\pm h_3$ outside S_2 has $Q = F_0$;*
(iii) *$M_3 = \mathbb{Z}g + \mathbb{Z}h + \mathbb{Z}h_3$ is primitive in K, with successive minima e_1, e_2, F_0 attained by g, h, h_3;*
(iv) *$|B(g,h_3)| < e_1$ and $|B(h,h_3)| < e_2$.*

[dep:C1C0-E]

Proof. Since S_2 is primitive, the number of vectors of K outside S_2 at each value n equals the number of vectors of L_j outside the plane of $S_{2,j}$ at n. For every catalogue vector of L_j below $9a$ outside that plane, the affine form $\tau - F_0$ was checked at all vertices of the two closures. The value $9a$ suffices, since F_0 is at most $7.8a$ on the closures. The outcome:

- $\tau - F_0 \geq 0$ everywhere, with exactly one pair where $\tau \equiv F_0$;
- every other tie lies on the excluded edges $\beta = 0$ or $\beta = \frac{1}{2}$, or at their vertices.

This gives (i) and (ii). For (iii), the vectors of M_3 below F_0 lie in S_2, whose successive minima are e_1, e_2, attained by g, h (Corollary 9.3); so M_3 has successive minima e_1, e_2, F_0, attained by g, h, h_3. If M_3 were not primitive, Lemma 5.5 would give z outside M_3, not in the plane of S_2, with $Q(z) \leq (e_1 + e_2 + F_0)/4$. This is less than F_0, because $3F_0 - e_1 - e_2 > 0$ on both closures (the margin is at least $5a$), contradicting (i). Part (iv): $h_3 \pm g$ and $h_3 \pm h$ lie outside S_2 and are not $\pm h_3$. □

The known $M_{3,j} \subset L_j$ is C2's, up to the sign of h_3. A vector v shortest in a coset that generates K/M_3 gives $K = M_3 \oplus \mathbb{Z}v$ with

$$F_0 < f = Q(v) \leq \frac{\Delta}{2 \det M_3} + \frac{e_1 + e_2 + F_0}{4},$$

and the smaller pair norm $t_i = \min_{\pm} Q(v \pm s_i)$, $s_i \in \{g, h, h_3\}$, satisfies $t_i \leq f + F_0$. Exact rational interval bounds on a cover of each closure (by 5,150 and 11,900 cells) give the windows of Table 5.

chamber	graph	$f_{\max}$, known $M_{3,j}$	window, known $M_{3,j}$	$f_{\max}$, any M_3	window, any M_3
C1	$0, 2$	$9.061a$	$14.638a$	$13.233a$	$18.463a$
C1	1	$9.444a$	$15.944a$	$12.141a$	$18.266a$
C0	$0, 2$	$10.343a$	$16.852a$	$13.521a$	$19.991a$
C0	1	$10.717a$	$18.517a$	$12.679a$	$20.020a$

TABLE 5. Upper bounds for the fourth basis vector and its smaller pair norms, rounded up to three decimals. For an arbitrary M_3, Minkowski's bound $\det M_3 \geq 4e_1e_2F_0$ (Lemma 7.8, with Lemma 9.5(iii)) is used.

9.4. **The rank-three span.**

Proposition 9.6 (Stage 1)**.** *For an arbitrary mate with $S_2 \cong S_{2,j}$, the Gram matrix of M_3 is that of $M_{3,j}$, except at isolated points and on one segment. Every isolated point has $a \leq 10$ or $D \not\equiv 2 \pmod 3$, except $(15, 2, 46)$, which lies on the segment and carries its alternative. The segment lies in* C1*, for graph* 2*:*

$$c = 2a + 8b, \qquad \tfrac{4}{33} \leq \beta \leq \tfrac{1}{7},$$

with the alternative pairing $B_{12} = c-a-3b$, $B_{13} = 2c-5a-4b$, $B_{23} = 4b-c$. *Its endpoint* $(\frac{1}{7}, \frac{22}{7})$ *lies on the wall* $c = 3a+b$. *It contains infinitely many in-scope seeds with* $a > 10$, *such as* $(15, 2, 46)$ *and* $(22, 3, 68)$. *[dep:C1C0-ST1]*

Proof outline. The symbolic classification of Proposition 8.3 was run on the closures of the two chambers, at windows $17a$ (C1) and $21a$ (C0), box radius 3, with generator norms (e_1, e_2, F_0) and the single pair at F_0 enforced by the counts. The known $M_{3,j}$ survives on the whole closure in every graph. An exact per-seed enumeration at every primitive seed with $a \leq 24$, for all three graphs (9,414 rows), lies inside the symbolic prediction everywhere and finds $M_{3,j}$ at every seed. It keeps the segment alternative at its in-scope seeds with $a \leq 24$, which are $(15, 2, 46)$, $(22, 3, 68)$ and $(23, 3, 70)$.

9.5. **The completion.** *Case I: completing* $M_{3,j}$. The branch universe is that of C2's Case I (Section 8.3), on the closed triangle and the closed quadrilateral, with $f \leq 10a$ on C1 and $f \leq 11a$ on C0, and with the catalogue below $16a$ (C1) and $19a$ (C0) for the target values. These windows contain those of Table 5 for the known $M_{3,j}$.

Case II: the segment. The alternative M_3 of Proposition 9.6 is completed on the whole closed segment, with $f \leq 14a$ and window $19a$.

Remark 9.7 (The certificate box). The zero-free certificates of decision N (Section 5.4) subdivide a fixed box $\beta \in [0, \frac{1}{2}]$, $\gamma \in [G_0, G_0 + 2]$. The closed domain must lie inside the box: zeros outside the box are never examined. The original box, $G_0 = 1$, stops at $\gamma = 3$, but C1 reaches $\gamma = \frac{7}{2}$ and C0 reaches $\frac{24}{5}$. These runs therefore use $G_0 = 2$ ($\gamma \in [2, 4]$) for C1 and the segment, and $G_0 = 3$ ($\gamma \in [3, 5]$) for C0, with the same algorithm and resolution. Before every run, a guard checks exactly that the region cut out of the configured box by the half-planes equals the configured domain [`dep:CERTBOX`]. The validations of the configurable box were:

(a) with $G_0 = 1$ it reproduces the sealed C4 pilot exactly;
(b) it agrees with the fixed box on random cases inside it, and correctly refuses polynomials with zeros above $\gamma = 3$ that the fixed box certifies wrongly;
(c) at a larger window ($19a$) on C2's Case I, the predeclared row-level equality of the theta-matching candidates failed (376 against 377). The completion-level conclusions agree: no unresolved branch, every match isometric to some L_j, and all controls pass. An additional analysis, not predeclared, shows that each of the seven exchanged rows is the same completion, found as a point record in one run and inside an identified family piece in the other. The failed row-level gate is kept on record, together with this explanation.

The candidates.

- On C1 there are 114 distinct candidates. Of these, 110 agree with θ_L through $16a$, and each is isometric, by an exact isometry test, to

run	branches	N	R	O	L	P	K	U
C1, Case I	1,022,040	958,971	60,987	1,884	107	127	72	0
C0, Case I	869,052	826,215	40,017	2,524	143	207	111	0
C1, the segment	276,972	273,870	3,102	—	—	—	—	0

TABLE 6. The C1 and C0 completions. A branch can carry several letters [`dep:UNIVERSES`].

the L_j of its own span (72 copies of L_1, 19 of L_0, 19 of L_2). The other 4 are at $(7, 2, 22)$, with $D \equiv 0 \pmod 3$, and differ from θ_L first between $6.1a$ and $6.6a$.

- On C0 all 156 distinct candidates agree with θ_L through $16a$, and each is isometric, by an exact isometry test, to the L_j of its own span (91, 39 and 26 copies of L_1, L_2, L_0).
- On the segment no completion exists: every branch is N or R.

Controls, all predeclared and all passing:

- **K1.** L_j is found as an identified K piece at every primitive seed of the chamber with $a \le 40$ (4,444 seeds of C1, 10,204 of C0), for every graph. An additional check at $a \le 100$ (69,378 and 161,527 seeds) finds none missing. This control was executed by a separate script on the correct seed universe; the engine's own seed generator for it covers only $c \le 3a$.
- **K2** and **K2′**. Brute force equals the symbolic hits at 30, 30 and 5 control pairs, and every theta-matching brute-force completion is found (44, 41, and 0 on the segment, where none exists).
- **The segment.** At its seeds $(15, 2, 46)$, $(22, 3, 68)$ and at the endpoint seed $(7, 1, 22)$, among others, brute force finds no completion, in agreement with the symbolic result.

The segment, checked exhaustively. An independent portable consumer checks every one of the 276,972 branches of the segment, both endpoints included, with closed rational-root proofs for the R decisions; no branch is nonroutine [`dep:C1-SEG`]. An earlier exhaustive check of the same branches (260,538 strict Bernstein certificates, 15,960 nonzero point checks and 474 exact rational-root exclusions) is kept as historical support.

Remark 9.8 (What an R decision means). An R decision is a statement about the eligible *generic* rational points of its domain: $0 < \beta < \frac{1}{2}$, with the chamber's strict inequalities. It is not a claim that the determinant has no rational solution on the closed domain. That stronger reading is false. Here is an R branch of C0:

- graph 0, $f = 9a$, $t_1 = 4a + 2b + c$, $t_2 = 7a + 4b + c$, $t_3 = 3a + 3b + c$, all signs $+$;

- at the non-generic seed $(2,1,9)$, where $\beta=\frac{1}{2}$, its Gram matrix is

$$\begin{pmatrix} 20 & 7 & -4 & 9 \\ 7 & 24 & 5 & 3 \\ -4 & 5 & 24 & 12 \\ 9 & 3 & 12 & 36 \end{pmatrix},$$

positive definite, with determinant $210{,}681 = 729 \cdot 17^2 = (27D)^2$.

This lattice is in fact integrally isometric to $L_0(2,1,9)$, by an explicit unimodular map [dep:PORTABLE-C0] (Remark 11.1). It lies outside Theorem 9.1 because $2b = a$, and gives no fourth generic class. A solver-free checker confirmed all 866,232 routine (N and R) decisions of C0's Case I from exact certificates, rebuilding the branch index independently. For twelve R branches the last piece lies on a non-generic boundary: eleven cuts on $\beta=\frac{1}{2}$ and one whole domain at $(\beta,\gamma)=(0,3)$. These pieces are certified by an exact witness that they lie outside the generic chamber. That witness can use only the strict conditions $\beta>0$, $\beta<\frac{1}{2}$ and $\gamma>3+\beta$, never the included tail wall $20c=81a+30b$ [dep:PORTABLE-C0]. See also Remark 11.1.

9.6. **Proof of Theorem 9.1.** Let K be a theta mate at a seed of the theorem.

(1) By Corollary 9.3, K contains the primitive binary $S_2=\mathbb{Z}g+\mathbb{Z}h$, and by Theorem 9.4, $S_2\cong S_{2,j}$.
(2) By Lemma 9.5, $M_3=\mathbb{Z}g+\mathbb{Z}h+\mathbb{Z}h_3$ is primitive. By Proposition 9.6, M_3 is $M_{3,j}$, or the seed lies on the segment, since every other isolated alternative point has $a\le 10$ or $D\not\equiv 2 \pmod 3$.
(3) Then $K=M_3\oplus\mathbb{Z}v$ with v in the branch universe of Case I or of the segment. Every branch is decided with $U=0$ (Table 6). Every K piece is an explicit L_j; every candidate is isometric to the L_j of its span or fails θ_L at a seed with $D\not\equiv 2 \pmod 3$; and the segment has no completion.

So K is isometric to some L_j. [dep:C1C0-ST2] □

10. Common scalings and the proof of the main theorem

Lemma 10.1 (Common scalings)**.** *Let $(a,b,c)=k(a_0,b_0,c_0)$ with $k\ge 1$ and (a_0,b_0,c_0) primitive. Suppose the main theorem holds for (a_0,b_0,c_0). Then it holds for (a,b,c).*

Proof. The Gram matrices of the construction at (a,b,c) are k times those at (a_0,b_0,c_0). Let K be a theta mate of $L_0(a,b,c)$.

- Every value $Q_K(x)$ is a value of Q on $L_0(a,b,c)$, so it is divisible by k.
- By polarization, $B(x,y)=Q(x+y)-Q(x)-Q(y)$ is divisible by k as well.

Hence the Gram matrix of K divided by k is integral and even. Its lattice K' is positive definite and has the theta series of $L_0(a_0,b_0,c_0)$. By the

primitive theorem, $K' \cong L_j(a_0, b_0, c_0)$ for exactly one j. Rescaling gives $K \cong L_j(a, b, c)$. □

For the same reason the residue conditions of Section 3 are applied to the primitive seed, after the common content is removed. The main theorem itself needs no such condition.

Proof of the main theorem. Theorem A supplies the three theta mates L_0, L_1, L_2, and Theorem B shows that they are pairwise non-isometric for every generic seed [`dep:THM-AB`]. It remains to show that every theta mate is isometric to one of them.

- For primitive seeds with $D \not\equiv 2 \pmod 3$, this is Section 3.
- For primitive seeds with $D \equiv 2 \pmod 3$, the chambers of Table 1 cover $\gamma > 1$, with every wall assigned (Remark 4.1). Each row is proved in the section the table names, except finitely many seeds with $a \leq 10$, which are closed by the finite register (Remark 4.2).
- Common scalings follow from Lemma 10.1.

□

11. Computation and verification

11.1. **The branch universes.** The new completion searches run over six finite symbolic branch universes (Table 7). A branch fixes a combinatorial type of basis: the span data, the norms and the pairings of the fourth vector. Its domain is an exact closed polygon, segment or point in (β, γ). Every branch receives a decision. A branch may carry several labels when its domain contains several pieces or points.

- **N** (no zero): the determinant condition has no real zero on the whole closed domain, proved by an exact certificate.
- **R:** no rational completion at an eligible generic point survives the determinant and the necessary theta tests.
- **K** (known): the branch is identified with an explicit integral basis of a known L_j.
- **P** (points): finitely many rational points are settled by exact theta differences and exact isometry.
- **Families:** identity families are excluded by count tests on the whole family; such branches are recorded as R.
- **O, L:** pieces outside the open domain, and pieces failing the value test, are recorded as such.

In every universe the number of unresolved branches is $U = 0$.

Remark 11.1 (R decisions are generic statements). An R decision concerns the generic chamber (Section 5): $0 < \beta < \frac{1}{2}$, the chamber's walls included. It does not assert anything at non-generic boundary points. This distinction is real. At the non-generic seed $(2, 1, 9)$, where $\beta = \frac{1}{2}$, one R branch of C0 has an exact positive-definite determinant solution. That solution is in fact

Universe	Branches	Unresolved
C4	6,611,512	0
C3, graphs 0 and 2	3,223,584	0
C3, graph 1 (both cells and the wall)	15,584,164	0
C1, known spans	1,022,040	0
C0 below the tail, known spans	869,052	0
C1, exceptional segment	276,972	0

TABLE 7. The six completion universes [dep:UNIVERSES].

integrally isometric to $L_0(2,1,9)$, a known theta mate. It lies outside the main theorem because $2b = a$, and it gives no fourth generic class. Included walls of the theorem, such as $20c = 81a+30b$, are never treated as boundaries in this sense [dep:PORTABLE-C0].

11.2. **Engines, domains and family exclusions.** The searches use the completion engines recorded in [dep:ENG]. Each run is configured by an exact chamber domain. Two rules matter for soundness.

- *The certificate box.* The zero-free certificates are proved on a fixed box in (β, γ), which must contain the actual closed domain. A guard checks this containment before any production run. The C1 and C0 runs use $\gamma \in [2,4]$ and $\gamma \in [3,5]$. An earlier fixed box $\gamma \leq 3$ would not have covered them, and it is not used there [dep:CERTBOX].
- *Family points.* A list of rational points attached to a family is not a list of every point passing an earlier value test. A whole-family count test may reject a point by a stronger criterion, so a checker must validate the family exclusion itself.

11.3. **What has been checked independently.** We state each check exactly.

- *Portable consumers of every routine decision.* For each of the six completion universes, a separate solver-free checker (standard-library arithmetic only, written independently of the production engines) rebuilds the complete branch index from the chamber data and consumes an exact certificate for every routine decision, in source order, with no missing or extra record. Table 8 gives the counts. Literal N decisions are closed real-domain statements; R decisions concern the generic chamber with its walls included (Remark 11.1). The certificates are produced by separate emitters, which are not trusted: only the consumers carry the check.
- *Nonroutine branches.* The 68,689 branch presentations that the routine certificate language does not decide (and the 2,820 of C0) keep the earlier independent audit: it rebuilds the complete branch universe and checks every nonroutine branch, every known map and every candidate key by exact computation. The audits rely on the

SMT solver z3 (release 5.3.3 on both platforms), whose exact decisions are trusted (no independently checked solver trace is supplied). Its nonlinear zero-existence queries, with linear queries where a restriction vanishes, settle the 35,539 nonroutine presentations covered by rational curves (about half of the 71,509) and the predeclared sample of routine decisions. Every nonroutine query was decided within the declared time limit; the few sampled routine queries that reached it are closed by exact Bernstein certificates. On a much slower machine a query could reach the limit, and that job would then fail closed.

- *The exceptional* C1 *segment.* Its portable consumer checks all 276,972 branches, with both endpoints included and closed rational-root proofs for the R decisions; no branch is nonroutine. The earlier exhaustive check of the same segment (with 260,538 strict Bernstein certificates, 15,960 nonzero point checks and 474 exact rational-root exclusions) is kept as historical support [`dep:C1-SEG`].
- *Replays.* The jobs of the earlier accepted manifest were re-executed in full on Windows, each in a fresh directory holding only its declared inputs, and every final output reproduced. That manifest did not give every proof step an executed route: audits found nine steps without one, namely the tail rungs up to $W = 57/5$ (the rung at $37/4$ and the windows from $19/2$ to $57/5$), the certificate ladder above $57/5$, the binary-span recognition of Theorem 8.1, the two seed-span enumerations, the scope link from Proposition 7.11 to Proposition 7.12, the finite checks of Theorems A and B, and the within-seed certificate; their jobs are among those added for this release. The jobs added for this release were run by their authors and partly re-executed, in independent reviews and in selected verifier runs. The runs of record are complete runs of both manifests (release and regeneration) on the frozen release bundle, in the environment described under Reproduction. Their logs and outcomes, with the archive, manifest and verifier hashes, are published with the evidence of the release, outside the bundle.

For C0, 866,220 routine decisions are certified by the original certificate types; for the remaining 12 R branches the last piece lies on a non-generic boundary (eleven cuts on $\beta = \frac{1}{2}$, one whole domain at $(\beta, \gamma) = (0, 3)$), and is certified by an exact outside-the-generic-domain witness [`dep:PORTABLE-C0`].

The finite register. The seeds with $a \leq 10$ that the chamber arguments set aside (Remark 4.2) are the 228 primitive rows with $D \equiv 2 \pmod 3$ below the tail. For each, a certificate file records the complete p-neighbour closure of the genus of $M \otimes A_2$ at one admissible prime (with an explicit basis and isometry for every neighbour), a one-coefficient exclusion for every discarded class, and, for every retained class, all 81 radical complements,

Universe	Routine, checked	Nonroutine, audited
C4	6,578,693	32,819
C3, graphs 0 and 2	3,211,339	12,245
C3, graph 1, lower cell	4,165,542	5,446
C3, graph 1, upper cell	4,114,007	6,161
C3, graph 1, wall $2c = 2a + 3b$	7,283,072	9,936
C1, known spans	1,019,958	2,082
C0 below the tail, known spans	866,232	2,820
C1, exceptional segment	276,972	0
Total	27,515,815	71,509

TABLE 8. Routine decisions consumed by the portable checkers, and nonroutine branch presentations covered by the independent audit. Counts are of branch presentations: a branch whose domain meets a wall is counted in each cell.

each either theta-different from L_0 at an explicit coefficient or integrally isometric to some L_t by an explicit basis. An independent consumer checks all of them: 79,929 classes, 3,074,496 neighbour certificates, 392 retained classes and 31,752 complement certificates. Completeness of the closure is the accepted single-spinor-genus and neighbour-connectivity argument; the certificates do not reprove it [`dep:REG`].

Checksums establish that the data are unchanged. They do not check any mathematical decision.

11.4. **Reproduction.** From the root of the accompanying data:

```
cd results/rank4-tensor-proof-package/v2
python -B verify_r18.py quick
python -B verify_r18.py full --mirror-root DIR
python -B verify_r18.py full --deps REGENERATION_MANIFEST.json --mirror-root DIR
```

- `quick` re-hashes every sealed and pinned file, checks each step's recorded mathematical final status, and checks that the coverage map gives every computational step an executable route. It takes minutes and reruns no mathematics.
- `full` re-executes every job of the manifest.
  - Each job runs in a fresh directory holding only its declared inputs, verified by hash, under a fail-closed memory guard on its whole process tree.
  - A job passes only if its exit code is the declared one, every final-status predicate holds, and every declared output equals the sealed one under its comparison rule. A zero exit code alone is not accepted.
  - Only a run of the whole manifest is reported as a complete verification; a run of selected jobs is labelled as a selection.

- The coverage map classifies every step as a hand proof, a computational dependency, a supporting check or a record, and every computational dependency names at least one required job.
- Without `--deps` the verifier uses the release manifest shipped beside it, which holds one necessary route per obligation: the certificate consumers, the validation suites, the nonroutine audits and the remaining checks. The regeneration manifest holds the producers of the certificate streams, the pilot runs, the controls and the historical outcomes kept as expected nonfinal results (never counted as completion). A complete run is always a run of the named manifest, and the closing line names it with its sha256.
- The supported environment for the runs of record is Linux, where every sealed file runs unchanged.
  - A small compatibility layer, installed as the interpreter's `sitecustomize` module and pinned by its hash in the manifest, reproduces Windows behaviour that some scripts rely on: text-mode line endings, the default text encoding, the start method of process pools (spawn, as on Windows), and the priority, CPU-affinity and memory calls that some scripts make.
  - The nonroutine audits load a Linux build of the same z3 release; it is the one file added on Linux.
  - Nineteen scripts name a fixed Windows path for PARI/GP. On Linux that path, read as a file name, must resolve on the job's search path to PARI/GP 2.17; otherwise the verifier refuses the run.
  - Every job runs under a portable process guard.
  - The runs of record start from archives extracted to the Linux file system, inside a private mount namespace without access to the Windows drives.

  The jobs were developed and first run on Windows. Two of them run on Linux only, because their deepest files exceed the Windows path-length limit inside a verifier mirror. The software versions are listed in the environment file of the data.

12. Related questions

The following are outside the main theorem. None is a dependency of it.

- *Fibres outside the family.* We prove no upper bound for rank-four theta fibres in general. The only statement we make about $\flat_4$ is $\flat_4 \geq 3$.
- *Non-generic parameters.* The cases $b = 0$, $2b = a$ and $a = c$ are not treated.

- *Sharper separation.* Uniform statements about where the theta series of different members of the family first differ remain open. They would sharpen, not complete, the main theorem.
- *Other constructions.* Theta-equal, non-isometric pairs also occur outside this family:
  - a two-parameter isospectral pencil of rank-four forms;
  - an isolated pair of level $3 \cdot 127$, found among the connecting ideals of an Eichler order of level 127 in the quaternion algebra ramified at 3 and ∞.

  In the corresponding pair-lattice sets at the nine prime levels $223, 229, 241, 263, 271, 283, 307, 313$ and 457, no theta-equal, non-isometric pair occurs. This is a computed, bounded result for those levels only; it has been re-classified independently and reviewed.

Appendix A. Dependencies

Each step of the proof has a stable identifier, cited in the text as `[dep:ID]`. The table gives its status and its verification route. Full paths, SHA-256 hashes, the final-status predicates and the job definitions are in the machine-readable supplement `SUPPLEMENT_MANIFEST.json`, read with the release manifest by the checker `verify_r18.py`. A *required* job checks a computation the step depends on, and belongs to the release manifest, the default reviewer command. A *regeneration* job reruns a producer whose sealed output a required job checks; a *supporting* job reruns a control, a regression or an earlier independent check; a *provenance* job reproduces a historical nonfinal outcome under its own contract. These three are in the regeneration manifest. An exit code alone is never accepted: every job must also reach its recorded mathematical final status and reproduce its outputs.

PROVED: a hand proof, accepted at review. *PROVED (computer-assisted)*: an exact computation is part of the proof. *COMPUTED*: the result of an exact computation. *CHECKED*: an independent verification of a claim. Rows marked *evidence only* are not dependencies.

ID	Step	Status; verification route
`THM-AB`	Theorems A and B (equal theta series; pairwise non-isometry)	proved (the earlier release's finite checks, rerun by its verifier); 1 required job.
`RES-R1`	Global ambient rigidity R1	proved (computer-assisted); 2 required jobs.
`RES-D1`	Congruence reduction for $D \equiv 1 \pmod 3$	proved (computer-assisted: an exact finite certificate); 2 required jobs.

ID	Step	Status; verification route
RES-D0	Two visible descents for $D \equiv 0, 3 \pmod 9$	proved (with a finite residue certificate); 1 required job.
RES-D6	$D \equiv 6 \pmod 9$: uniform ascents and (R_Z)	proved (computer-assisted); 19 required jobs, 4 supporting.
LADDER	The tail ladder above $W = 57/5$	proved (computer-assisted; every bundled module replayed in a copy of the bundle); 1 required job.
TAIL19	R(19/2): exact-three for $4c > 17a + 6b$	proved (computer-assisted; rerun in the archived work tree, restored unchanged; two jobs run on Linux only); 17 required jobs, 12 supporting.
TAIL	Tail theorem, $20c > 81a + 30b$	proved (computer-assisted; the rungs 37/4 and 91/10 here; the windows down to 19/2 in row TAIL19 and the ladder above 57/5 in row LADDER); 24 required jobs.
C4-REC	Complete $4a$-span recognition in C4	proved (computer-assisted); 8 required jobs, 6 supporting.
BASIS-BOUNDS	Bounds for the fourth basis vector	proved; hand proof; independent evidence listed in the supplement.
C4	The C4 completion (wall $c = a + b$ included)	proved (computer-assisted); 9 required jobs, 10 regeneration, 4 supporting, 2 provenance.
C3-REC-HYB	C3 recognition and the hybrid-exclusion theorem	proved (computer-assisted); 8 required jobs, 4 supporting.

ID	Step	Status; verification route
C3-02	The C3 completion, graphs 0 and 2	proved (computer-assisted); 8 required jobs, 7 regeneration, 1 supporting.
C3G1-A1	C3 graph 1: first excess and bounds	proved (computer-assisted: exact Bernstein certificates); 1 required job.
C3G1-A2	C3 graph 1: rank-three spans	computed (exact); 1 required job, 1 supporting.
C3G1-A3a	C3 graph 1: the alternative spans have no completion	computed (complete finite search); 2 required jobs.
C3G1-A3b	C3 graph 1: the completion on both cells and the tie wall	proved (computer-assisted); 25 required jobs, 15 regeneration, 4 supporting.
C3G1-A3c	C3 graph 1: end-to-end check for $a \leq 20$ (consistency only)	computed (not a dependency); evidence only; 1 job.
C2	The C2 completion, with the boundary repair	proved (computer-assisted); 11 required jobs, 3 supporting.
C1C0-S1	C1/C0: the first excess	computed (exact envelope); 2 required jobs.
C1C0-D	C1/C0: binary recognition	proved (computer-assisted); 5 required jobs, 4 supporting.
C1C0-E	C1/C0: Lemma E2′ and certified windows	proved (exact certificates); 2 required jobs, 2 supporting.
C1C0-ST1	C1/C0: the rank-three stage	computed (exact); 4 required jobs, 2 supporting.
C1C0-ST2	C1/C0: the completion with the certificate box	proved (computer-assisted); 9 required jobs, 8 regeneration, 1 supporting.

ID	Step	Status; verification route
`C1-SEG`	The exceptional C1 segment	proved (computer-assisted; a portable consumer checks every branch, both endpoints included); 2 required jobs, 6 regeneration, 4 supporting.
`CERTBOX`	The certificate-box domain guard	checked; 2 required jobs, 2 supporting.
`ENG`	The completion engines	sealed source; evidence only; 5 jobs.
`UNIVERSES`	The six completion universes	computed; record of accepted values, checked by the final-status predicates.
`PORTABLE-C0`	Solver-free certificates for every routine decision of C0 below the tail; the boundary isometry at $(2, 1, 9)$	checked; 4 required jobs, 3 regeneration.
`REG`	The finite register: the 228 primitive seeds with $a \leq 10$ and $D \equiv 2 \pmod 3$ below the tail	proved (computer-assisted; genus-closure certificates checked by an independent consumer); 3 required jobs, 1 regeneration, 4 supporting, 1 provenance.
`C4-200`	Bounded C4 census (152,994 seeds; evidence only)	computed (not a dependency); evidence only.
`AN`	The acceptance record	record; record of accepted values, checked by the final-status predicates.

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